Computer Networks

2nd Project Report

Class 14 – Group 6

Members

Guilherme Teixeira (up202204875)

Rafael Cunha (up202209857)

Summary

In the context of the Computer Networks FEUP UC, we were faced with the challenge of implementing a program that downloads files from a server using FTP, as well as building and implementing a functional computer network. Given the project, we obtained substantial knowledge in areas such as server interaction, as well as a better understanding of the network layer.

Introduction

In this report, we will document the various steps the project requires, such as:

1. Download Application
    - 1.1. Building the download application
    - 1.2. Testing the download application

2. Computer Network
    - 2.1. Configure an IP Network
    - 2.2. Implement two bridges in a switch
    - 2.3. Configure a Router in Linux
    - 2.4. Configure a Commercial Router and Implement NAT
    - 2.5. Using DNS
    - 2.6. TCP connections

Besides this, we will be explaining in detail the mechanisms of each step and answering the questions included in the 2nd lab guide, regarding the different concepts and technologies used.

1.1. Building the download application

The goal for this part of our work was to implement a command-line FTP client that enables users to download files from FTP servers. This application should support both anonymous and authenticated connection and, has the previous sentence suggests, follows the File Transfer Protocol (FTP) specified in class. The program we built handles URL parsing, network connections through socket management, server authentication, and file transfers through a passive data connection mode

In the following paragraphs, we will enumerate and explain each of the main implemented functionalities in detail. Then, we will provide an explanation of the overall application flow.

1.1.1 URL Parsing Component

URL decomposition is handled by the `parse` function through systematic regex pattern matching. It starts by validating the basic URL structure by checking for mandatory forward slashes. Then, it determines if authentication credentials are present by searching for the '@' symbol. For URLs without credentials, this function extracts the hostname and applies default anonymous FTP credentials (username: "anonymous", password: "password"). When we do have credentials, it parses the username and password using specific regex patterns. The function extracts the resource path and filename, with the filename being determined as the text following the last forward slash. Finally, it resolves the hostname to an IP address using DNS lookup through the `gethostbyname` method.

1.1.2 Socket Management

When it comes to Socket Management, `createSocket` function establishes network connections by initializing a TCP/IP socket structure. It starts by zero-ing out the server address structure using `bzero and converting the port number to network byte order using `htons` and the IP address using `inet_addr`. Then, it configures the socket with the AF_INET protocol family for IPv4 networking, creates a stream socket and attempts to establish a connection to the specified server. This entire process is done with error detection/handling for both socket creation and connection establishment failures.

1.1.3 Authentication System

The `authConn` function implements the FTP authentication protocol. Here, we start by dynamically allocating memory for command strings, constructing them with proper FTP command syntax including carriage return and newline characters. Then, we send the USER command and wait for the server to indicate readiness for the password (response code 331). When the server is ready, we send the PASS command and verify successful authentication. This process is done with memory management for allocated command buffers and comprehensive error detection/handling for both command transmission and server response validation.

1.1.4 Passive Mode Implementation

FTP passive mode negotiation is done through the `passiveMode` function. This function sends the PASV command and processes the server's response containing IP address and port information. The response format follows the FTP specification with comma-separated values enclosed in parentheses. Here, a total of six numbers is parsed: four for IP address octets and two for port calculation. The port number is computed by multiplying the first number by 256 and adding the second number. At the end, this function assembles the IP address string and validates the server's passive mode acceptance.

1.1.5 Resource Acquisition

To handle file downloads, we use the `requestResource` and `getResource` functions. In this context, `requestResource` constructs and sends the RETR command for the specified resource, declaring that the client wishes to download a copy of a file on the server. Then, `getResource` manages the actual file transfer, creating a local file and implementing a buffered read-write loop for data transfer. Here, error detection and handling was implemented for file operations and network transfers, tracking total bytes transferred and ensuring write operations complete successfully.

1.1.6 Response Processing

In the context of response processing, the `readResponse` function implements a state machine designed to handle FTP's varying response formats. Throughout this process, the response is accumulated in a buffer, with special attention to hyphen-prefixed lines that indicate continuation. After reaching the END state, the function extracts the numeric response code using a regex pattern. This state-based approach ensures robust parsing of FTP's complex response protocol while maintaining message integrity and preventing buffer overflow through careful boundary management.

1.1.7 Connection Closing

The `closeConnection` is a simple function that implements proper FTP session termination. It sends the QUIT command on the control connection and waits for the server's goodbye response. Here, we ensure both control and data connections are properly closed, preventing resource leaks. Error checking is included for the server's response and socket closure operations.

1.1.8 General Application Flow (`main` function)

The application begins by validating the command-line arguments and using the parse function to extract URL components: host, resource, file, user, password, and IP. Using the parsed IP and FTP port 21, createSocket establishes the control connection (socketA).

Authentication is managed by the `authConn` function, which verifies the username and password via socketA. The passiveMode function then enables passive mode, processing server responses to determine the IP and port for the data connection, and creates socketB with these parameters.

File transfer starts with requestResource, which sends the RETR command through socketA, and getResource, which handles the file transfer using both socketA and socketB. Throughout, readResponse monitors and processes server responses on socketA, ensuring operations succeed. Finally, closeConnection cleans up socketA and socketB before termination.

1.1.9 Report of a successful download

Using the test case `./download ftp://anonymous:anonymous@ftp.bit.nl/speedtest/100mb.bin`, we trace the application's operation: The **URL parser** identifies ftp.bit.nl (host), speedtest/100mb.bin (resource), 100mb.bin (file), "anonymous" (credentials), and resolves the IP to 213.136.12.213. **Socket management** creates socketA on port 21, enabling the **authentication system** to send USER/PASS commands. The **response processor** handles the server's multi-line 230 responses through its state machine.

The **passive mode** component processes the PASV response (227), calculates port 51083 from (199,139), and establishes socketB for data transfer. **Resource acquisition** initiates with requestResource sending RETR through socketA, while getResource manages the 100MB transfer through socketB. The **response processor** monitors status updates until completion (226), when the **connection manager** cleanly closes both sockets after the goodbye message (221).

The image below shows the completed transfer process.

```
rscscp@DESKTOP-C9F06I5:/mnt/c/Users/Rafa/Desktop/RCOMp2$ ./download ftp://anonymous:anonymous@ftp.bit.nl/speedtest/100mb.bin
Host: ftp.bit.nl
Resource: speedtest/100mb.bin
File: 100mb.bin
User: anonymous
Password: anonymous
IP Address: 213.136.12.213
220 Welcome to ftp.bit.nl
Socket to '213.136.12.213' and port 21 succeeded
331 Anonymous login ok, send your complete email address as your password
Username successful
Password successful
230-
230-Welcome to our *PUBLIC* OPEN SOURCE SOFTWARE MIRROR SERVER.
230-Please DO NOT report this under our responsible disclosure policy.
230-This is a PUBLIC service, with OPEN SOURCE SOFTWARE, and NOT a security threat to our company.
230-There is NO SENSITIVE INFORMATION on this server.
230-
230-Thanks.
230-
230-
230 Anonymous access granted, restrictions apply
Authentication succeeded with username = 'anonymous' and password = 'anonymous'.
227 Entering Passive Mode (213,136,12,213,199,139).
Passive mode succeeded
Socket to '213.136.12.213:51083' success
150 Opening ASCII mode data connection for speedtest/100mb.bin (104857600 bytes)
Transfer ready to begin
226 Transfer complete
221 Goodbye.
rscscp@DESKTOP-C9F06I5:/mnt/c/Users/Rafa/Desktop/RCOMp2$
```

1.2. Testing the download application

We tested the application on our pc's using public tests outside the FEUP FTP server, and during these tests we verified that our program went through 3 phases. First, it went through a series of responses/requests where it entered the server, created the socket, logged the user using the credentials given, entered passive mode, and opened the file. During this phase we can see the packets interchanging between FTP packets with ACK and SYN messages, where the server and the user acknowledge one another and give a response/request.

During the exchange of data, we have TCP packets that manage the data being transferred, using the TCP congestion control to increase or decrease the limit of packets waiting to be transferred. The data itself is included in FTP-Data packets, which including informations as the PASV code, which symbolizes we are in Passive Mode. In the middle of the transfer we can see where the transfer went wrong, by the presence of duplicate ACK, which are the way the client informs the server the packets were not received correctly, where the server follows by retransmitting said packets.

Once the data transfer is finished, we can see ARP messages related to the MAC of the sending computer, followed by the final responses and requests, which close the connection.

2.1. Configure an IP Network

In this step, the goal is to establish a connection between 2 computers within the same network (172.16.Y0.0). For this we firstly connect ports E1 of tuxY3 and E1 of tuxY4 to ports 3 and 4 of the switch, respectively. Afterwards we configure eth1interface of tuxY3 using the command *ifconfig eth1 172.16.Y0.1/24*, which defines the IP of tuxY3's eth1 interface as 172.16.Y0.1, as well as defining its bitmask as 255.255.255.0 (the MAC is 00:c0:df:25:40:66). We do the same for tux64's eth1interface, using *ifconfig eth1 172.16.Y0.254/24* (the MAC is 00:01:02:a1:35:69). These IP's are used to connections within the network layer, while the MAC's are used for communication inside the data link layer. ARP packets are created in order to map IP's to MAC's, allowing for a smooth transition between the logical address (IP) and physical address (MAC). Each ARP packet includes a sender IP address and MAC address, as well as a target IP address and MAC address (set to 00:00:00:00:00:00 initially, as it's unknown in the beginning).

Next, we use the command *ping 172.16.Y0.254* to try and establish a connection from tuxY3 to tuxY4. The connection was successful, and the 2 tuxes interchanged 2 types of packets: ICMP Echo Request (tuxY3 to tuxY4) and ICMP Echo Reply (tuxY4 to tuxY3). Since we are making a ping request within the same network, the only addresses in it are the sender and destination IP and MAC addresses. The main difference between these addresses and the ones in ARP packets is that while in ARP packets the destination MAC is unknown, in ping packets this MAC is already known, thanks to the ARP tables.

The next step is to analyze the routes and arp lists, using *route –n and arp –a* respectively. In the route part, both tuxes include routes to 10.227.20.254 in eth0, which were not created by us, and a route to 172.16.Y0.0 by 0.0.0.0 gateway in eth1, which means that both tuxes are directly connected to that network. In the arp part, each tux has a list of ip's it has connected, together with their MAC addresses. We then re-ping tux64, but use Wireshark to capture the packets being transmitted in eth1. Each of these packets are easily distinguishable in Wireshark, as they get properly named, however by just looking at an Ethernet frame, we can still distinguish them as ARP, IP and ICMP by their Ethertype value (which is a part of the header). If its value is 0x0800, it's IP, if it's 0x0806 it's ARP. ICMP can be checked if the IP header has value 1.

After capturing the packets, we can then verify that for every ICMP Request there is a corresponding ICMP Reply, where by seeing the IP's in the transactions, verifies that the connection between tux63 and tux64 was successful. We can also check aspects like the packet length in Wireshark, where by selecting a packet and checking its details, we can find useful aspects like the frame length. Each computer also has the possibility to use a loopback interface, which allows it to communicate with itself. This is extremely important for testing and software development, as it allows to check sent and received packets without another computer or network to interact.

2.2. Implement two bridges in a switch

The first step of this experience is to configure E1 of tux62, by connecting E1 of tuxY2 to port 2 of the switch and using the command *ifconfig eth1 172.16.Y1.1* (MAC is 00:e0:7d:b5:8c:8e). Then we connect tuxes in the same network with bridges, which means tuxY3 and tuxY4 belong to bridgeY0 and tuxY2 to bridgeY1. After creating both bridges using GTKTerm in tuxY3 (the switch console is connected to tuxY3's E0 port) (*/interface bridge add name=bridgeY0),* we remove the connections from ports 2,3 and 4 to the default bridge (/interface bridge port remove [find interface=ether{port}]) and connect them to their previously defined bridges (/interface bridge port add bridge=bridge{number} interface=ether{port}).

We then start the capture of packets in tuxY3.eth1, and ping tuxY4 and tuxY2. Since tuxY4 belongs to the same network and bridge as tuxY3 and tuxY2 doesn't, it is no shock to verify that pinging tuxY4 is a success, but pinging tuxY2 is a failure. After this process we start capturing tuxY2.eth1 and tuxY4.eth1 as well, then ping broadcast in tuxY3 (*ping -b 172.16.Y0.255*) [Annex A.14] and in tuxY2 [Annex A.13]. Each network has its own broadcast, and in the network shown in this experience we can see 3 different broadcast domains (172.16.Y0.254, 172.16.Y1.254, 10.227.20.255). In each broadcast every tux in the network is pinged, and that is why when the first broadcast is pinged, the request comes from tuxY3, but the reply comes from tuxY4. In the second case, both come from tuxY2, as it's the only connect to that network.

2.3. Configure a Router in Linux

In this experience, we will turn tuxY4 into a router, by following these simple steps. First, we configure E2 of tuxY4 (*ifconfig eth2 172.16.Y1.253*) (MAC is 00:c0:df:04:20:8c), connect tuxY4's E2 port to port 1 of the switch, and then add it to bridgeY1 (*/interface bridge port add bridge=bridgeY1* *interface=ether1*). Afterwards, we enable IP forwarding (*sysctl net.ipv4.ip_forward=1*) and finally we disable ICMP echo-ignore-broadcast (*sysctl net.ipv4.icmp_echo_ignore_broadcasts=0*). By doing this we allow the packets to be forwarded along various IP's, while having a forwarding table to keep track of it. A regular forwarding table entry will the next IP to be reached, the mask used, the gateway used (if exists), the interface used, the metric of this route (distance compared to routes with the same destination), and flags (usually UG, meaning route is up and has a gateway).

After having configured the router, we add new routes to tuxY3 (*route add –net 172.16.Y1.0/24 gw 172.16.Y0.254*) and tuxY2 (*route add –net 172.16.Y0.0/24 gw 172.16.Y1.253*) in order to connect them to the other network using tuxY4 as a gateway. Therefore, tuxY3 and tuxY2 both have a route to their own networks and a route to the other network by tuxY4. TuxY4 however only has 2 routes to its own networks, as eth1 and eth2 are connected to different networks, covering both.

We then capture packets at tuxY3, and from there ping tuxY4.eth1, tuxY4.eth2 and tuxY2 [Annex A.18]. During the first ping, we can see ARP messages being interchanged, where one tux tries to figure out the other's MAC address, where the MAC addresses of tuxY3 and tuxY4.eth1 are associated to these messages. For every connection, we can see ICMP request and reply packets being exchanged between the two tuxes communicating, as these symbolize a successful connection. The IP's associated to these are the IP's of the tuxes communicating, however when we analyze the MAC addresses present, we realize that evn when communicating with other tuxes, the MAC's used are always the ones of tuxY3 and tuxY4.eth1, which makes sense considering we are capturing in tuxY3, whose only known MAC address is tuxY4.eth1, which belongs to the same network.

This verifies itself afterwards, as the next step wants us to ping tuxY2 from tuxY3, while capturing at tuxY4.eth1 and tuxY4.eth2 [Annex A.19] [Annex A.20]. When we do so after cleaning the ARP tables in every tux we verify that, while in both captures the IP's mentioned are the ones of tuxY3 and tuxY2, in tuxY4.eth1 the MAC's associated with the packets are the ones from itself and tuxY3, while in tuxY4.eth2 the MAC's associated with the packets are the ones from itself and tuxY2 (which belong to the same network).

2.4. Configure a Commercial Router and Implement NAT

In order to begin this experience, we must configure the Microtik router. To do so, we must firstly use GTKTerm while still connected to the console to remove port 5 from its connection to the default bridge and connect it to bridgeY1. Then, connect E1 Rc to port PY.12 of the lab network and E2 Rc to port 5 of the Microtik console. Then we simply use GTKTerm, which now serves as the router serial console, and configure the IP addresses for ether1 (*/ip address add address=172.16.1.Y9/24 interface=ether1*) and ether2 (*/ip address add address=172.16.Y1.254/24 interface=ether2*). Then we must add a route to 172.16.1.0/24 for tuxY3 (route add –net 172.16.1.0/24 gw 172.16.Y0.254), tuxY4 and tuxY2 (route add –net 172.16.1.0/24 gw 172.16.Y1.254), as well as a static route to 172.16.Y0.0/24 for Rc (/ip route add dst-address=172.16.Y0.0/24 gateway=172.16.Y1.253). By starting the capture at tuxY3 and pinging tuxY2, tuxY4 and Rc, we can see by the ICMP Requests and Replies that all connections are successful. [Annex A.1]

Afterwards we go to tuxY2 and disable the ability to accept ICMP Redirect messages for eth1 and all network interfaces (*sysctl net.ipv4.conf.eth1.accept_redirects=0 / sysctl net.ipv4.conf.all.accept_redirect=0*). We then alter the routes so that Rc is the new gateway to 172.16.Y0.0/24, instead of tuxY4.eth2 (*route del -net 172.16.Y0.0/24 gw 172.16.Y1.253/ route add -net 172.16.Y0.0/24 gw 172.16.Y1.254*). After pinging tuxY3, we can see that the ICMP messages include the regular replies and requests between tuxY2 and tuxY3, but also redirect messages from Rc to tuxY2 [Annex A.2]. This indicates that the path followed is tuxY2-> Rc-> tuxY4->tuxY3. This happens because the only direct way tuxY2 has of reaching the 172.16.Y0.0 network is by Rc, and Rc itself needs to go through tuxY4 to reach said network. When we reinstate tuxY4.eth2 as the gateway to the route from tuxY2 to 172.16.Y0.0, the path returns to being tuxY2->tuxY4->tuxY3 [Annex A.3]. Finally, when we use Rc as the gateway, but allow redirect messages, after the first exchanged a redirect occurs, skipping Rc in the route between tuxY3 and tuxY2 [Annex A.4].

In the final stage, we successfully ping the FTP server (172.16.1.10) from tuxY3 [Annex A.5]. This is possible because of the existence of NAT (Network Address Translation). NAT maps the origin IP address of the packets to the router's IP, allowing it to communicate with the FTP server. Without it, the FTP server can't return the packets to tuxY3, for example, as it doesn't know its location. By using NAT, the server can drop the packets in the router, which can then lead them to the correct destination. We prove this by disabling NAT (*/ip firewall nat disable 0*) and retrying to ping the FTP server, which results in ICMP Requests from tuxY3 which are not replied by the server [Annex A.6]. NAT can be reinstated using the command */ip firewall nat add chain=srcnat action=masquerade out-interface=ether1*.

2.5. Using DNS

In this experience, we will be using DNS, and to configure it in all tuxes, we simply have to access their resolv.conf file (*nano /etc/resolv.conf*), and add/replace if existent the line *nameserver 10.227.20.3.* In order to test it, we can ping an online browser, like *ping google.com*. [Annex A.7] [Annex A.8] [Annex A.9] When successful, the packets exchanged by DNS are DNS Queries, where the host sends the hostname to be resolved (google.com in our example), and DNS Responses, where the DNS server sends the IP address corresponding to the requested hostname.

2.6. TCP connections

For this step, we only need to compile our application and start a capture in tuxY3 and run our application [Annex A.10]. After this, we repeat this step, however while the program is running, we start a capture and run our application on tuxY2 [Annex A.11] [Annex A.12]. Each one can be divided in 3 phases (establishment, data and termination).

During the first connection, during the establishment phase of the TCP control, we can see an interchange of TCP and FTP packets. The TCP packets can be divided into 3 groups: SYN (packets where the client initiates the connection), SYN-ACK (the server acknowledges the requests and responds) and ACK (the client acknowledges the response). These responses and requests are sent through FTP packets using the FTP control connection.

During the data phase, the TCP ARQ Mechanism is used to deliver data reliably over a network. The data is divided, and each segment has a sequence number assigned to it. Whenever these segments reach the receiver, it sends an ACK packet with the sequence number of the next expected byte. Whenever a sender doesn't receiver an ACK within a certain timeout, it simply retransmits the segment assuming it was lost or corrupted. On the other side, if a receiver doesn't receive a segment but receives the ones following, it sends a duplicate ACK of the last received byte, which the server uses to resend the lost segment.

In order to manage the amount of data waiting to be sent, the TCP congestion control is implemented. There are some techniques implemented like slow start, where the sender starts with a small congestion window in order to figure out the limit of the bandwidth. This window keeps increasing with each ACK received, until a packet loss occurs. After this, instead of growing exponentially, the window grows linearly, which decreases the possibility of overshooting the limit capacity. In case of errors, detected by duplicate ACK's or surpassing the timeout, the capacity is severely limited and, after retransmitting the packet, the program starts from the slow start point. This can be seen in our program, as the connection keeps increasing exponentially until a certain point, and whenever a duplicate ACK happens, this flow severely diminishes.

In the termination phase, we can see ARP messages regarding the MAC's of tuxY3, and a final response/request phase similar to the establishment phase.

The FTP application opens 2 TCP connections, one used for request and responses, where the FTP control information is transported, and another for transferring data.

During our second connection, in the moment our second application is running we can see an abundance in duplicate ACK, as the bandwidth available cannot handle the current capacity of tuxY3 alongside tuxY2 capacity. Therefore, the capacity of tuxY3 is adapted during this phase so that both applications can run smoothly. Eventually, the flow is resumed, which is both due to the fact that tuxY3 eventually reached a safe limit to avoid congestion, and that tuxY2 download eventually stopped.

Conclusion

During this project, we were able to develop our knowledge of every step of a network build, including terms like bridges, routers, etc... Besides this, we were able to learn more about the relations between servers and clients, as well as the connection between the network layer and the data link layer.

Annexes

Annex A.

A.1 Exp4 Step3 Tux3 Wireshark

1	0.000000000	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
2	2.002073199	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
3	4.004120383	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
4	6.006188484	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
5	8.008248240	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
6	10.010342812	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
7	11.153333307	172.16.90.1	172.16.90.254	ICMP	98	Echo (ping) request id=0x171f, seq=1/256, ttl=64 (reply in 8)
8	11.153491915	172.16.90.254	172.16.90.1	ICMP	98	Echo (ping) reply id=0x171f, seq=1/256, ttl=64 (request in 7)
9	12.012447964	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
10	12.174239421	172.16.90.1	172.16.90.254	ICMP	98	Echo (ping) request id=0x171f, seq=2/512, ttl=64 (reply in 11)
11	12.174375471	172.16.90.254	172.16.90.1	ICMP	98	Echo (ping) reply id=0x171f, seq=2/512, ttl=64 (request in 10)
12	13.198246884	172.16.90.1	172.16.90.254	ICMP	98	Echo (ping) request id=0x171f, seq=3/768, ttl=64 (reply in 13)
13	13.198370711	172.16.90.254	172.16.90.1	ICMP	98	Echo (ping) reply id=0x171f, seq=3/768, ttl=64 (request in 12)
14	14.014539392	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
15	16.016605363	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
16	16.180047996	KYE_25:40:81	KYE_08:d5:9a	ARP	60	Who has 172.16.90.1? Tell 172.16.90.254
17	16.180066644	KYE_08:d5:9a	KYE_25:40:81	ARP	42	172.16.90.1 is at 00:c0:df:08:d5:9a
18	16.366209825	KYE_08:d5:9a	KYE_25:40:81	ARP	42	Who has 172.16.90.254? Tell 172.16.90.1
19	16.366311303	KYE_25:40:81	KYE_08:d5:9a	ARP	60	172.16.90.254 is at 00:c0:df:25:40:81
20	17.993316740	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=1/256, ttl=64 (reply in 21)
21	17.993883007	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=1/256, ttl=63 (request in 20)
22	18.018691063	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
23	18.438437359	0.0.0.0	255.255.255.255	MNDP	159	5678 → 5678 Len=117
24	18.438467041	Routerboardc_1c:95:c9	CDP/VTP/DTP/PAgP/UDLD	CDP	93	Device ID: MikroTik Port ID: bridge90
25	18.438516628	Routerboardc_1c:95:c9	LLDP_Multicast	LLDP	110	MA/c4:ad:34:1c:95:c9 IN/bridge90 120 SysN=MikroTik SysD=MikroTik RouterOS 6.43.16 (long-term) CRS326-24G-2S+
26	19.022241586	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=2/512, ttl=64 (reply in 27)
27	19.022833764	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=2/512, ttl=63 (request in 26)
28	20.020802988	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
29	20.046242753	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=3/768, ttl=64 (reply in 30)
30	20.046583086	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=3/768, ttl=63 (request in 29)
31	21.070239948	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=4/1024, ttl=64 (reply in 32)
32	21.070601373	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=4/1024, ttl=63 (request in 31)
33	22.022935621	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
34	22.094243158	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=5/1280, ttl=64 (reply in 35)
35	22.094656684	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=5/1280, ttl=63 (request in 34)
36	23.118245330	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=6/1536, ttl=64 (reply in 37)
37	23.118499829	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=6/1536, ttl=63 (request in 36)
38	24.025042725	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
39	24.142242342	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=7/1792, ttl=64 (reply in 40)
40	24.142854704	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=7/1792, ttl=63 (request in 39)
41	25.166244252	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) request id=0x1727, seq=8/2048, ttl=64 (reply in 42)
42	25.166720425	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1727, seq=8/2048, ttl=63 (request in 41)
43	26.027119343	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
44	27.608962179	172.16.90.1	172.16.91.254	ICMP	98	Echo (ping) request id=0x172e, seq=1/256, ttl=64 (reply in 45)
45	27.609415864	172.16.91.254	172.16.90.1	ICMP	98	Echo (ping) reply id=0x172e, seq=1/256, ttl=63 (request in 44)
46	28.029239226	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
47	28.622244714	172.16.90.1	172.16.91.254	ICMP	98	Echo (ping) request id=0x172e, seq=2/512, ttl=64 (reply in 48)
48	28.622503822	172.16.91.254	172.16.90.1	ICMP	98	Echo (ping) reply id=0x172e, seq=2/512, ttl=63 (request in 47)
49	29.646240376	172.16.90.1	172.16.91.254	ICMP	98	Echo (ping) request id=0x172e, seq=3/768, ttl=64 (reply in 50)
50	29.646485377	172.16.91.254	172.16.90.1	ICMP	98	Echo (ping) reply id=0x172e, seq=3/768, ttl=63 (request in 49)
51	30.031304248	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
52	32.032806878	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
53	34.034901719	Routerboardc_1c:95:c9	Spanning-tree-(for-bridge…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001

A.2 Exp4 Step4 Tux2 Part1 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
2	2.002190872	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
3	4.004398482	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
4	6.006399470	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
5	8.008295529	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
6	8.876297729	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=1/256, ttl=64 (no response found!)
7	8.876635281	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=1/256, ttl=63 (reply in 8)
8	8.876854797	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=1/256, ttl=63 (request in 7)
9	9.893957670	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=2/512, ttl=64 (reply in 13)
10	9.894480235	Routerboardc_eb:24:1d	Broadcast	ARP	60	Who has 172.16.91.1? Tell 172.16.91.254
11	9.894492038	EdimaxTechno_59:be:3a	Routerboardc_eb:24:1d	ARP	42	172.16.91.1 is at 00:50:fc:59:be:3a
12	9.894612168	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
13	9.894704291	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=2/512, ttl=63 (request in 9)
14	10.010269367	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
15	10.917967239	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=3/768, ttl=64 (reply in 17)
16	10.918201353	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
17	10.918426596	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=3/768, ttl=63 (request in 15)
18	11.941960109	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=4/1024, ttl=64 (reply in 20)
19	11.942209658	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
20	11.942411016	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=4/1024, ttl=63 (request in 18)
21	12.012608598	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
22	12.965938865	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=5/1280, ttl=64 (reply in 24)
23	12.966506688	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
24	12.966720129	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=5/1280, ttl=63 (request in 22)
25	13.989920477	EdimaxTechno_59:be:3a	Routerboardc_eb:24:1d	ARP	42	Who has 172.16.91.254? Tell 172.16.91.1
26	13.989945761	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=6/1536, ttl=64 (reply in 29)
27	13.990413639	Routerboardc_eb:24:1d	EdimaxTechno_59:be:3a	ARP	60	172.16.91.254 is at 74:4d:28:eb:24:1d
28	13.990426560	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
29	13.990616742	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=6/1536, ttl=63 (request in 26)
30	14.004405494	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
31	14.059179160	KYE_08:d5:99	EdimaxTechno_59:be:3a	ARP	60	Who has 172.16.91.1? Tell 172.16.91.253
32	14.059184817	EdimaxTechno_59:be:3a	KYE_08:d5:99	ARP	42	172.16.91.1 is at 00:50:fc:59:be:3a
33	15.013953418	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=7/1792, ttl=64 (reply in 34)
34	15.014656457	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=7/1792, ttl=63 (request in 33)
35	16.006733920	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
36	16.037953735	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=8/2048, ttl=64 (reply in 38)
37	16.038145454	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
38	16.038358545	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=8/2048, ttl=63 (request in 36)
39	17.061938122	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=9/2304, ttl=64 (reply in 40)
40	17.062810880	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=9/2304, ttl=63 (request in 39)
41	18.008651853	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
42	18.085952115	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x11ef, seq=10/2560, ttl=64 (reply in 43)
43	18.086464623	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x11ef, seq=10/2560, ttl=63 (request in 42)
44	18.653148480	192.168.88.1	255.255.255.255	MNDP	153	5678 → 5678 Len=111
45	18.653165731	Routerboardc_eb:24:1d	CDP/VTP/DTP/PAgP/UDLD	CDP	108	Device ID: MikroTik Port ID: bridge
46	20.010619140	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
47	22.012709117	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
48	24.014828123	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
49	26.017028053	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002

A.3 Exp4 Step4 Tux2 Part2 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
2	1.871299611	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x1279, seq=1/256, ttl=64 (reply in 3)
3	1.871733405	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x1279, seq=1/256, ttl=63 (request in 2)
4	2.002170445	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
5	2.892647761	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x1279, seq=2/512, ttl=64 (reply in 6)
6	2.893023377	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x1279, seq=2/512, ttl=63 (request in 5)
7	3.916644281	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x1279, seq=3/768, ttl=64 (reply in 8)
8	3.917007325	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x1279, seq=3/768, ttl=63 (request in 7)
9	4.004250636	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
10	4.940644568	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x1279, seq=4/1024, ttl=64 (reply in 11)
11	4.940991548	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x1279, seq=4/1024, ttl=63 (request in 10)
12	5.964644151	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x1279, seq=5/1280, ttl=64 (reply in 13)
13	5.964962705	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x1279, seq=5/1280, ttl=63 (request in 12)
14	6.006337583	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
15	6.988614676	EdimaxTechno_59:be:3a	KYE_08:d5:99	ARP	42	Who has 172.16.91.253? Tell 172.16.91.1
16	6.988652531	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x1279, seq=6/1536, ttl=64 (reply in 18)
17	6.988811703	KYE_08:d5:99	EdimaxTechno_59:be:3a	ARP	60	172.16.91.253 is at 00:c0:df:08:d5:99
18	6.988954671	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x1279, seq=6/1536, ttl=63 (request in 16)
19	7.060393365	KYE_08:d5:99	EdimaxTechno_59:be:3a	ARP	60	Who has 172.16.91.1? Tell 172.16.91.253
20	7.060411944	EdimaxTechno_59:be:3a	KYE_08:d5:99	ARP	42	172.16.91.1 is at 00:50:fc:59:be:3a
21	8.008611831	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002

A.4 Exp4 Step4 Tux2 Part3 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
2	0.425268623	192.168.88.1	255.255.255.255	MNDP	153	5678 → 5678 Len=111
3	0.425283918	Routerboardc_eb:24:1d	CDP/VTP/DTP/PAgP/UDLD	CDP	108	Device ID: MikroTik Port ID: bridge
4	2.002072884	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
5	2.683344100	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=1/256, ttl=64 (reply in 6)
6	2.683817216	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=1/256, ttl=63 (request in 5)
7	3.694878307	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=2/512, ttl=64 (reply in 9)
8	3.695093702	172.16.91.254	172.16.91.1	ICMP	126	Redirect (Redirect for host)
9	3.695301136	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=2/512, ttl=63 (request in 7)
10	4.004200510	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
11	4.718876953	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=3/768, ttl=64 (reply in 12)
12	4.719214713	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=3/768, ttl=63 (request in 11)
13	5.742871125	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=4/1024, ttl=64 (reply in 14)
14	5.743224111	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=4/1024, ttl=63 (request in 13)
15	6.006421222	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
16	6.766875350	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=5/1280, ttl=64 (reply in 17)
17	6.767441427	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=5/1280, ttl=63 (request in 16)
18	7.735634745	KYE_08:d5:99	EdimaxTechno_59:be:3a	ARP	60	Who has 172.16.91.1? Tell 172.16.91.253
19	7.735653323	EdimaxTechno_59:be:3a	KYE_08:d5:99	ARP	42	172.16.91.1 is at 00:50:fc:59:be:3a
20	7.790868676	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=6/1536, ttl=64 (reply in 21)
21	7.791670613	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=6/1536, ttl=63 (request in 20)
22	7.918839800	EdimaxTechno_59:be:3a	Routerboardc_eb:24:1d	ARP	42	Who has 172.16.91.254? Tell 172.16.91.1
23	7.919699496	Routerboardc_eb:24:1d	EdimaxTechno_59:be:3a	ARP	60	172.16.91.254 is at 74:4d:28:eb:24:1d
24	8.008877218	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
25	8.814873802	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=7/1792, ttl=64 (reply in 26)
26	8.815695504	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=7/1792, ttl=63 (request in 25)
27	9.838873126	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=8/2048, ttl=64 (reply in 28)
28	9.839506951	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=8/2048, ttl=63 (request in 27)
29	9.966834331	EdimaxTechno_59:be:3a	KYE_08:d5:99	ARP	42	Who has 172.16.91.253? Tell 172.16.91.1
30	9.967464663	KYE_08:d5:99	EdimaxTechno_59:be:3a	ARP	60	172.16.91.253 is at 00:c0:df:08:d5:99
31	10.010950111	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
32	10.862874961	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=9/2304, ttl=64 (reply in 33)
33	10.863311060	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=9/2304, ttl=63 (request in 32)
34	11.886872602	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=10/2560, ttl=64 (reply in 35)
35	11.887336358	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=10/2560, ttl=63 (request in 34)
36	12.012887773	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002
37	12.910872333	172.16.91.1	172.16.90.1	ICMP	98	Echo (ping) request id=0x12b3, seq=11/2816, ttl=64 (reply in 38)
38	12.911265269	172.16.90.1	172.16.91.1	ICMP	98	Echo (ping) reply id=0x12b3, seq=11/2816, ttl=63 (request in 37)
39	14.014480605	Routerboardc_1c:95:d3	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:1d Cost = 10 Port = 0x8002

A.5 Exp4 Step5 Tux3 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
2	1.278598082	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1910, seq=1/256, ttl=64 (reply in 3)
3	1.279122795	172.16.1.10	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1910, seq=1/256, ttl=62 (request in 2)
4	2.002164290	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
5	2.305046321	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1910, seq=2/512, ttl=64 (reply in 6)
6	2.305468019	172.16.1.10	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1910, seq=2/512, ttl=62 (request in 5)
7	3.329041476	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1910, seq=3/768, ttl=64 (reply in 8)
8	3.329450812	172.16.1.10	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1910, seq=3/768, ttl=62 (request in 7)
9	4.004247854	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
10	4.353041033	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1910, seq=4/1024, ttl=64 (reply in 11)
11	4.353432491	172.16.1.10	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1910, seq=4/1024, ttl=62 (request in 10)
12	5.377041431	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1910, seq=5/1280, ttl=64 (reply in 13)
13	5.377416127	172.16.1.10	172.16.90.1	ICMP	98	Echo (ping) reply id=0x1910, seq=5/1280, ttl=62 (request in 12)
14	6.006394982	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
15	6.305005625	KYE_08:d5:9a	KYE_25:40:81	ARP	42	Who has 172.16.90.254? Tell 172.16.90.1
16	6.305106964	KYE_25:40:81	KYE_08:d5:9a	ARP	60	172.16.90.254 is at 00:c0:df:25:40:81
17	6.387865116	KYE_25:40:81	KYE_08:d5:9a	ARP	60	Who has 172.16.90.1? Tell 172.16.90.254
18	6.387879643	KYE_08:d5:9a	KYE_25:40:81	ARP	42	172.16.90.1 is at 00:c0:df:08:d5:9a
19	8.008544286	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001

A.6 Exp4 Step7 Tux3 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
2	1.380953596	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=1/256, ttl=64 (no response found!)
3	2.002125513	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
4	2.392284274	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=2/512, ttl=64 (no response found!)
5	3.416283093	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=3/768, ttl=64 (no response found!)
6	4.004216532	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
7	4.440288618	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=4/1024, ttl=64 (no response found!)
8	5.464290094	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=5/1280, ttl=64 (no response found!)
9	6.006336053	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
10	6.488249528	KYE_08:d5:9a	KYE_25:40:81	ARP	42	Who has 172.16.90.254? Tell 172.16.90.1
11	6.488284239	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=6/1536, ttl=64 (no response found!)
12	6.488360994	KYE_25:40:81	KYE_08:d5:9a	ARP	60	172.16.90.254 is at 00:c0:df:25:40:81
13	7.512282367	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=7/1792, ttl=64 (no response found!)
14	8.008486034	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
15	8.536284967	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=8/2048, ttl=64 (no response found!)
16	9.560280026	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=9/2304, ttl=64 (no response found!)
17	10.010627361	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
18	10.584282072	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=10/2560, ttl=64 (no response found!)
19	11.612280617	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=11/2816, ttl=64 (no response found!)
20	12.012796423	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001
21	12.632283095	172.16.90.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x1965, seq=12/3072, ttl=64 (no response found!)
22	14.014912554	Routerboardc_1c:95:c9	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:95:c9 Cost = 0 Port = 0x8001

A.7 Exp5 Step1 Tux2 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
2	0.181986570	Cisco_78:94:84	Spanning-tree-(for-bri…	STP	60	Conf. Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8004
3	0.327193228	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=6/1536, ttl=64 (reply in 4)
4	0.344230120	142.250.200.142	10.227.20.92	ICMP	98	Echo (ping) reply id=0x1510, seq=6/1536, ttl=112 (request in 3)
5	0.465221093	Routerboardc_20:25:c8	Broadcast	ARP	60	Who has 10.227.20.84? Tell 10.227.20.254
6	0.465377890	194.117.47.42	10.227.20.84	NTP	90	NTP Version 4, server
7	0.519164384	HewlettPacka_5a:7c:e7	Routerboardc_20:25:c8	ARP	42	Who has 10.227.20.254? Tell 10.227.20.92
8	0.519350165	Routerboardc_20:25:c8	HewlettPacka_5a:7c:e7	ARP	60	10.227.20.254 is at e4:8d:8c:20:25:c8
9	0.758720342	ProxmoxServe_e7:5e:5b	HewlettPacka_5a:7c:e7	ARP	60	Who has 10.227.20.92? Tell 10.227.20.3
10	0.758737313	HewlettPacka_5a:7c:e7	ProxmoxServe_e7:5e:5b	ARP	42	10.227.20.92 is at 00:21:5a:5a:7c:e7
11	0.797892440	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
12	0.797899005	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
13	0.797901450	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
14	0.797903754	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
15	0.797905989	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
16	0.908976353	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
17	0.908981382	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
18	0.908983617	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
19	0.908985922	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
20	0.908988227	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
21	1.328304795	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=7/1792, ttl=64 (reply in 22)
22	1.345208078	142.250.200.142	10.227.20.92	ICMP	98	Echo (ping) reply id=0x1510, seq=7/1792, ttl=112 (request in 21)
23	1.824120453	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
24	1.824136307	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
25	1.824138891	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
26	1.824141196	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
27	1.824143501	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
28	1.918922265	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
29	1.918936513	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
30	1.918939097	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
31	1.918941402	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
32	1.918943847	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
33	1.927228529	Cisco_78:94:84	Cisco_78:94:84	LOOP	60	Reply
34	2.199370899	Cisco_78:94:84	Spanning-tree-(for-bri…	STP	60	Conf. Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8004
35	2.330285136	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=8/2048, ttl=64 (reply in 36)
36	2.347215378	142.250.200.142	10.227.20.92	ICMP	98	Echo (ping) reply id=0x1510, seq=8/2048, ttl=112 (request in 35)
37	2.564476120	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
38	2.848110562	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
39	2.848127045	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
40	2.848129420	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
41	2.848132004	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
42	2.848134379	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
43	2.881283234	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
44	2.882681068	::	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
45	2.942860831	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
46	2.942868374	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
47	2.942870888	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
48	2.942873193	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
49	2.942875497	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113

No.	Time	Source	Destination	Protocol	Length	Info
49	2.942875497	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
50	2.966964556	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
51	3.210443374	Cisco_78:94:84	CDP/VTP/DTP/PAgP/UDLD	DTP	60	Dynamic Trunk Protocol
52	3.210541643	Cisco_78:94:84	CDP/VTP/DTP/PAgP/UDLD	DTP	90	Dynamic Trunk Protocol
53	3.328849605	Cisco_78:94:84	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8004
54	3.331291170	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=9/2304, ttl=64 (reply in 56)
55	3.339769456	10.227.20.3	10.227.20.122	DNS	129	Standard query response 0xafd1 A 1.debian.pool.ntp.org A 23.147.168.176 A 193.136.152.72 A 91.209.16.78
56	3.348195290	142.250.200.142	10.227.20.92	ICMP	98	Echo (ping) reply id=0x1510, seq=9/2304, ttl=112 (request in 54)
57	3.582823629	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
58	3.882502637	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
59	3.968096795	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
60	3.974836484	10.227.20.158	255.255.255.255	DNS	83	Standard query 0xa7a9 A pool.ntp.org OPT
61	4.152785163	10.227.20.3	10.227.20.102	DNS	122	Standard query response 0x3d9c PTR 103.20.227.10.in-addr.arpa PTR tux103.netlab.fe.up.pt
62	4.333268995	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=10/2560, ttl=64 (reply in 64)
63	4.334638402	Cisco_78:94:84	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8004
64	4.350232272	142.250.200.142	10.227.20.92	ICMP	98	Echo (ping) reply id=0x1510, seq=10/2560, ttl=112 (request in 62)
65	4.606768898	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
66	4.906517190	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
67	4.911138897	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
68	4.911156567	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
69	4.911159780	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
70	4.911162155	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
71	4.911164739	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
72	4.953987729	10.227.20.3	10.227.20.189	DNS	131	Standard query response 0x245d PTR 26.112.82.140.in-addr.arpa PTR lb-140-82-112-26-iad.github.com
73	4.954489340	10.227.20.3	10.227.20.189	DNS	138	Standard query response 0xb59c PTR 93.243.107.34.in-addr.arpa PTR 93.243.107.34.bc.googleusercontent.com
74	4.992094376	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
75	5.000429694	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
76	5.000434373	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
77	5.000436748	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
78	5.000438913	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
79	5.000441218	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
80	5.334307504	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=11/2816, ttl=64 (reply in 81)
81	5.351252063	142.250.200.142	10.227.20.92	ICMP	98	Echo (ping) reply id=0x1510, seq=11/2816, ttl=112 (request in 80)
82	5.693529586	Routerboardc_20:25:c8	HewlettPacka_19:02:42	ARP	60	10.227.20.254 is at e4:8d:8c:20:25:c8
83	5.918701141	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
84	5.918716716	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
85	5.918719091	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
86	5.918721605	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
87	5.918723840	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
88	6.016102223	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
89	6.016108229	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
90	6.016110604	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
91	6.016112839	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
92	6.016115213	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
93	6.220076560	10.227.20.3	10.227.20.113	DNS	131	Standard query response 0x6b95 PTR 25.112.82.140.in-addr.arpa PTR lb-140-82-112-25-iad.github.com
94	6.220539687	10.227.20.3	10.227.20.113	DNS	138	Standard query response 0x5119 PTR 93.243.107.34.in-addr.arpa PTR 93.243.107.34.bc.googleusercontent.com
95	6.225669221	Cisco_78:94:84	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8004
96	6.335254797	10.227.20.92	142.250.200.142	ICMP	98	Echo (ping) request id=0x1510, seq=12/3072, ttl=64 (no response found!)

A.8 Exp5 Step1 Tux3 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
2	0.150007029	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
3	0.150024978	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
4	0.150027352	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
5	0.150029727	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
6	0.150031892	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
7	0.178088483	Cisco_78:94:88	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8008
8	0.258871771	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
9	0.258877847	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
10	0.258880012	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
11	0.258882386	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
12	0.258884551	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
13	0.302998753	10.227.20.93	142.250.200.142	ICMP	98	Echo (ping) request id=0x1c0f, seq=4/1024, ttl=64 (reply in 15)
14	0.305492051	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
15	0.321077750	142.250.200.142	10.227.20.93	ICMP	98	Echo (ping) reply id=0x1c0f, seq=4/1024, ttl=112 (request in 13)
16	0.389870158	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
17	0.441266956	Routerboardc_20:25:c8	HewlettPacka_61:2c:54	ARP	60	Who has 10.227.20.93? Tell 10.227.20.254
18	0.441277991	HewlettPacka_61:2c:54	Routerboardc_20:25:c8	ARP	42	10.227.20.93 is at 00:21:5a:61:2c:54
19	0.461359799	142.250.201.74	10.227.20.93	TLSv1.2	150	Application Data
20	0.461376281	10.227.20.93	142.250.201.74	TCP	66	37110 → 443 [ACK] Seq=1 Ack=85 Win=501 Len=0 TSval=1130478392 TSecr=3216030167
21	1.157874451	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
22	1.157890095	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
23	1.157892679	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
24	1.157894984	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
25	1.157897149	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
26	1.279920991	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
27	1.279930000	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
28	1.279932305	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
29	1.279934540	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
30	1.279936984	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
31	1.304146912	10.227.20.93	142.250.200.142	ICMP	98	Echo (ping) request id=0x1c0f, seq=5/1280, ttl=64 (reply in 32)
32	1.320946366	142.250.200.142	10.227.20.93	ICMP	98	Echo (ping) reply id=0x1c0f, seq=5/1280, ttl=112 (request in 31)
33	1.515246042	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x9be0484b
34	2.181828619	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
35	2.181844403	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
36	2.181846847	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
37	2.181849082	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
38	2.181851247	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
39	2.195559919	Cisco_78:94:88	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8008
40	2.238986544	fe80::221:5aff:fe61:2f…	ff02::2	ICMPv6	70	Router Solicitation from 00:21:5a:61:2f:d6
41	2.303855465	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
42	2.303872716	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
43	2.303875230	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
44	2.303877465	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
45	2.303879700	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
46	2.306019466	10.227.20.93	142.250.200.142	ICMP	98	Echo (ping) request id=0x1c0f, seq=6/1536, ttl=64 (reply in 47)
47	2.324916713	142.250.200.142	10.227.20.93	ICMP	98	Echo (ping) reply id=0x1c0f, seq=6/1536, ttl=112 (request in 46)
48	2.588394600	ProxmoxServe_e7:5e:5b	HewlettPacka_61:2c:54	ARP	60	Who has 10.227.20.93? Tell 10.227.20.3
49	2.588415412	HewlettPacka_61:2c:54	ProxmoxServe_e7:5e:5b	ARP	42	10.227.20.93 is at 00:21:5a:61:2c:54

No.	Time	Source	Destination	Protocol	Length	Info
50	2.715495624	10.227.20.2	10.227.20.35	DHCP	343	DHCP ACK - Transaction ID 0x8f05aaa8
51	2.949582620	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
52	2.952510535	10.227.20.2	10.227.20.125	DHCP	344	DHCP ACK - Transaction ID 0xd309cae1
53	3.275497590	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
54	3.307985759	10.227.20.93	142.250.200.142	ICMP	98	Echo (ping) request id=0x1c0f, seq=7/1792, ttl=64 (reply in 55)
55	3.324909179	142.250.200.142	10.227.20.93	ICMP	98	Echo (ping) reply id=0x1c0f, seq=7/1792, ttl=112 (request in 54)
56	3.348876553	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
57	3.823849084	10.227.20.93	10.227.20.3	DNS	85	Standard query 0x0b36 PTR 5.17.217.172.in-addr.arpa
58	3.824380778	10.227.20.3	10.227.20.93	DNS	123	Standard query response 0x0b36 PTR 5.17.217.172.in-addr.arpa PTR mad07s09-in-f5.1e100.net
59	3.824558661	10.227.20.93	10.227.20.3	DNS	88	Standard query 0x9497 PTR 174.184.250.142.in-addr.arpa
60	3.824931399	10.227.20.3	10.227.20.93	DNS	127	Standard query response 0x9497 PTR 174.184.250.142.in-addr.arpa PTR mad07s23-in-f14.1e100.net
61	3.825068984	10.227.20.93	10.227.20.3	DNS	87	Standard query 0xcc17 PTR 163.215.58.216.in-addr.arpa
62	3.825375583	10.227.20.3	10.227.20.93	DNS	125	Standard query response 0xcc17 PTR 163.215.58.216.in-addr.arpa PTR mad41s07-in-f3.1e100.net
63	3.825506883	10.227.20.93	10.227.20.3	DNS	87	Standard query 0x21d0 PTR 74.201.250.142.in-addr.arpa
64	3.825858249	10.227.20.3	10.227.20.93	DNS	126	Standard query response 0x21d0 PTR 74.201.250.142.in-addr.arpa PTR mad07s25-in-f10.1e100.net
65	3.825988362	10.227.20.93	10.227.20.3	DNS	87	Standard query 0x202f PTR 74.201.250.142.in-addr.arpa
66	3.826344478	10.227.20.3	10.227.20.93	DNS	126	Standard query response 0x202f PTR 74.201.250.142.in-addr.arpa PTR mad07s25-in-f10.1e100.net
67	3.967754034	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
68	4.216929120	Cisco_78:94:88	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8008
69	4.261122383	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
70	4.261129646	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
71	4.261132230	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
72	4.261134465	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
73	4.261136560	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
74	4.305532709	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
75	4.309977206	10.227.20.93	142.250.200.142	ICMP	98	Echo (ping) request id=0x1c0f, seq=8/2048, ttl=64 (reply in 76)
76	4.326894550	142.250.200.142	10.227.20.93	ICMP	98	Echo (ping) reply id=0x1c0f, seq=8/2048, ttl=112 (request in 75)
77	4.352501285	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
78	4.352505755	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
79	4.352507920	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
80	4.352510574	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
81	4.352512739	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
82	4.357855242	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
83	4.991702230	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
84	5.279674931	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
85	5.279681007	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
86	5.279683591	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
87	5.279685756	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
88	5.279688130	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
89	5.311966709	10.227.20.93	142.250.200.142	ICMP	98	Echo (ping) request id=0x1c0f, seq=9/2304, ttl=64 (reply in 90)
90	5.328883984	142.250.200.142	10.227.20.93	ICMP	98	Echo (ping) reply id=0x1c0f, seq=9/2304, ttl=112 (request in 89)
91	5.329541391	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
92	5.381866090	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
93	5.381870280	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
94	5.381872375	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
95	5.381874540	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
96	5.381876705	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
97	5.381891651	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123

A.9 Exp5 Step1 Tux4 Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
2	0.260561583	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
3	0.260567589	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
4	0.260569964	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
5	0.260572199	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
6	0.260574294	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
7	0.344995688	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
8	0.399843267	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
9	0.539802560	10.227.20.183	142.250.200.142	ICMP	98	Echo (ping) request id=0x1464, seq=5/1280, ttl=64 (reply in 10)
10	0.556759294	142.250.200.142	10.227.20.183	ICMP	98	Echo (ping) reply id=0x1464, seq=5/1280, ttl=112 (request in 9)
11	0.996091608	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78e40c66
12	1.023939411	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
13	1.075638708	Cisco_78:94:85	Spanning-tree-(for-bri…	STP	60	Conf. Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8005
14	1.155329263	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
15	1.155347492	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
16	1.155349937	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
17	1.155352102	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
18	1.155354476	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
19	1.279936451	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
20	1.279941270	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
21	1.279943435	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
22	1.279945880	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
23	1.279948115	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
24	1.369017304	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
25	1.419796429	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
26	1.531684217	HewlettPacka_19:09:5c	ProxmoxServe_e7:5e:5b	ARP	42	Who has 10.227.20.3? Tell 10.227.20.183
27	1.532096985	ProxmoxServe_e7:5e:5b	HewlettPacka_19:09:5c	ARP	60	10.227.20.3 is at bc:24:11:e7:5e:5b
28	1.540821545	10.227.20.183	142.250.200.142	ICMP	98	Echo (ping) request id=0x1464, seq=6/1536, ttl=64 (reply in 29)
29	1.557796647	142.250.200.142	10.227.20.183	ICMP	98	Echo (ping) reply id=0x1464, seq=6/1536, ttl=112 (request in 28)
30	1.787683353	HewlettPacka_19:09:5c	Routerboardc_20:25:c8	ARP	42	Who has 10.227.20.254? Tell 10.227.20.183
31	1.787865991	Routerboardc_20:25:c8	HewlettPacka_19:09:5c	ARP	60	10.227.20.254 is at e4:8d:8c:20:25:c8
32	1.954224349	ProxmoxServe_e7:5e:5b	HewlettPacka_19:09:5c	ARP	60	Who has 10.227.20.183? Tell 10.227.20.3
33	1.954234057	HewlettPacka_19:09:5c	ProxmoxServe_e7:5e:5b	ARP	42	10.227.20.183 is at 00:22:64:19:09:5c
34	2.012685289	fe80::221:5aff:fe5a:7d…	ff02::2	ICMPv6	70	Router Solicitation from 00:21:5a:5a:7d:12
35	2.155809976	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
36	2.155825900	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
37	2.155828414	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
38	2.155830789	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
39	2.155832884	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
40	2.303882781	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
41	2.303887460	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
42	2.303889905	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
43	2.303892000	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
44	2.303894025	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
45	2.541881182	10.227.20.183	142.250.200.142	ICMP	98	Echo (ping) request id=0x1464, seq=7/1792, ttl=64 (reply in 46)
46	2.558875281	142.250.200.142	10.227.20.183	ICMP	98	Echo (ping) reply id=0x1464, seq=7/1792, ttl=112 (request in 45)
47	2.911370989	Cisco_78:94:85	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8005
48	3.179812882	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
49	3.179827968	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123

No.	Time	Source	Destination	Protocol	Length	Info
40	2.303882781	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
41	2.303887460	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
42	2.303889905	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
43	2.303892000	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
44	2.303894025	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
45	2.541881182	10.227.20.183	142.250.200.142	ICMP	98	Echo (ping) request id=0x1464, seq=7/1792, ttl=64 (reply in 46)
46	2.558875281	142.250.200.142	10.227.20.183	ICMP	98	Echo (ping) reply id=0x1464, seq=7/1792, ttl=112 (request in 45)
47	2.911370989	Cisco_78:94:85	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8005
48	3.179812882	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
49	3.179827968	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
50	3.179829923	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
51	3.179831809	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
52	3.179833625	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
53	3.276572477	Cisco_78:94:85	Cisco_78:94:85	LOOP	60	Reply
54	3.528563296	Cisco_b6:8c:01	Cisco_b6:8c:01	LOOP	60	Reply
55	3.542962544	10.227.20.183	142.250.200.142	ICMP	98	Echo (ping) request id=0x1464, seq=8/2048, ttl=64 (reply in 56)
56	3.559777567	142.250.200.142	10.227.20.183	ICMP	98	Echo (ping) reply id=0x1464, seq=8/2048, ttl=112 (request in 55)
57	3.913968558	Cisco_78:94:85	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8005
58	3.970177574	10.227.20.3	10.227.20.103	DNS	122	Standard query response 0xa153 PTR 102.20.227.10.in-addr.arpa PTR tux102.netlab.fe.up.pt
59	3.970577211	10.227.20.3	10.227.20.103	DNS	122	Standard query response 0xc3c2 PTR 104.20.227.10.in-addr.arpa PTR tux104.netlab.fe.up.pt
60	3.993441905	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
61	4.262859217	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
62	4.262875560	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
63	4.262877795	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
64	4.262880169	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
65	4.262882404	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
66	4.330982770	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
67	4.393406843	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123
68	4.397539552	10.227.20.3	10.227.20.122	DNS	131	Standard query response 0xfa4e PTR 26.112.82.140.in-addr.arpa PTR lb-140-82-112-26-iad.github.com
69	4.397939818	10.227.20.3	10.227.20.122	DNS	138	Standard query response 0x1ebb PTR 93.243.107.34.in-addr.arpa PTR 93.243.107.34.bc.googleusercontent.com
70	4.443578481	Cisco_7c:9c:88	Cisco_7c:9c:88	LOOP	60	Reply
71	4.544860995	10.227.20.183	142.250.200.142	ICMP	98	Echo (ping) request id=0x1464, seq=9/2304, ttl=64 (reply in 72)
72	4.561801874	142.250.200.142	10.227.20.183	ICMP	98	Echo (ping) reply id=0x1464, seq=9/2304, ttl=112 (request in 71)
73	5.023739705	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.113
74	5.102356087	Cisco_78:94:85	Spanning-tree-(for-bri…	STP	60	Conf. TC + Root = 32768/0/4c:00:82:2e:9a:00 Cost = 19 Port = 0x8005
75	5.279727460	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.111? Tell 192.168.109.113
76	5.279742197	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.112? Tell 192.168.109.113
77	5.279744711	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.114? Tell 192.168.109.113
78	5.279746736	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.115? Tell 192.168.109.113
79	5.279748901	ASUSTekCOMPU_b3:e9:e8	Broadcast	ARP	60	Who has 192.168.109.116? Tell 192.168.109.113
80	5.337091039	HewlettPacka_61:2f:24	Broadcast	ARP	60	Who has 10.227.20.3? Tell 10.227.20.14
81	5.357796558	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.126? Tell 192.168.109.123
82	5.357801167	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.125? Tell 192.168.109.123
83	5.357803333	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.124? Tell 192.168.109.123
84	5.357805428	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.122? Tell 192.168.109.123
85	5.357807593	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.121? Tell 192.168.109.123
86	5.400544028	10.227.20.3	10.227.20.102	DNS	122	Standard query response 0x3bbc PTR 103.20.227.10.in-addr.arpa PTR tux103.netlab.fe.up.pt
87	5.419825253	ASUSTekCOMPU_2e:20:7c	Broadcast	ARP	60	Who has 192.168.109.1? Tell 192.168.109.123

A.10 Exp6 Step3 Tux3

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
2	2.002074588	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
3	4.004174944	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
4	6.006281511	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
5	8.008384930	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
6	10.010455728	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
7	12.012551036	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
8	14.014647177	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
9	16.016763568	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
10	18.018866405	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
11	18.862907400	172.16.110.1	172.16.1.10	TCP	74	51166 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3610997869 TSecr=0 WS=128
12	18.863418853	172.16.1.10	172.16.110.1	TCP	74	21 → 51166 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395708354 TSecr=3610997869 WS=128
13	18.863438688	172.16.110.1	172.16.1.10	TCP	66	51166 → 21 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3610997869 TSecr=3395708354
14	18.869297103	172.16.1.10	172.16.110.1	FTP	116	Response: 220 ProFTPD Server (Debian) [::ffff:172.16.1.10]
15	18.869307649	172.16.110.1	172.16.1.10	TCP	66	51166 → 21 [ACK] Seq=1 Ack=51 Win=64256 Len=0 TSval=3610997875 TSecr=3395708360
16	18.869422818	172.16.110.1	172.16.1.10	FTP	77	Request: user rcom
17	18.869679418	172.16.1.10	172.16.110.1	TCP	66	21 → 51166 [ACK] Seq=51 Ack=12 Win=65280 Len=0 TSval=3395708360 TSecr=3610997875
18	18.870379653	172.16.1.10	172.16.110.1	FTP	98	Response: 331 Password required for rcom
19	18.870451800	172.16.110.1	172.16.1.10	FTP	77	Request: pass rcom
20	18.914055055	172.16.1.10	172.16.110.1	TCP	66	21 → 51166 [ACK] Seq=83 Ack=23 Win=65280 Len=0 TSval=3395708405 TSecr=3610997876
21	19.018158782	172.16.1.10	172.16.110.1	FTP	113	Response: 230-Welcome, archive user rcom@172.16.1.119 !
22	19.018179944	172.16.1.10	172.16.110.1	FTP	115	Response:
23	19.018246504	172.16.1.10	172.16.110.1	FTP	142	Response:
24	19.018295253	172.16.1.10	172.16.110.1	FTP	157	Response: please report them via e-mail to <root@ftp.netlab.fe.up.pt>.
25	19.018606330	172.16.110.1	172.16.1.10	TCP	66	51166 → 21 [ACK] Seq=23 Ack=346 Win=64128 Len=0 TSval=3610998025 TSecr=3395708509
26	19.018672191	172.16.110.1	172.16.1.10	FTP	74	Request: type I
27	19.018977260	172.16.1.10	172.16.110.1	TCP	66	21 → 51166 [ACK] Seq=346 Ack=31 Win=65280 Len=0 TSval=3395708510 TSecr=3610998025
28	19.019259212	172.16.1.10	172.16.110.1	FTP	85	Response: 200 Type set to I
29	19.019315086	172.16.110.1	172.16.1.10	FTP	72	Request: pasv
30	19.020089564	172.16.1.10	172.16.110.1	FTP	116	Response: 227 Entering Passive Mode (172,16,1,10,151,251).
31	19.020216257	172.16.110.1	172.16.1.10	TCP	74	47424 → 38907 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3610998026 TSecr=0 WS=128
32	19.020566514	172.16.1.10	172.16.110.1	TCP	74	38907 → 47424 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395708511 TSecr=3610998026 WS=128
33	19.020581321	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3610998027 TSecr=3395708511
34	19.020621480	172.16.110.1	172.16.1.10	FTP	87	Request: retr files/crab.mp4
35	19.021511686	172.16.1.10	172.16.110.1	FTP	143	Response: 150 Opening BINARY mode data connection for files/crab.mp4 (29803194 bytes)
36	19.021979557	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
37	19.021988986	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 TSval=3610998028 TSecr=3395708512
38	19.022106390	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
39	19.022113794	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=2897 Win=63488 Len=0 TSval=3610998028 TSecr=3395708512
40	19.022224703	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41	19.022230709	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=4345 Win=62592 Len=0 TSval=3610998028 TSecr=3395708512
42	19.022348113	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
43	19.022353771	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=5793 Win=61568 Len=0 TSval=3610998028 TSecr=3395708512
44	19.022471035	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45	19.022477391	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=7241 Win=60672 Len=0 TSval=3610998028 TSecr=3395708512
46	19.022599056	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
47	19.022604922	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=8689 Win=59648 Len=0 TSval=3610998029 TSecr=3395708512
48	19.022718136	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
49	19.022724632	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=10137 Win=58752 Len=0 TSval=3610998029 TSecr=3395708512

No.	Time	Source	Destination	Protocol	Length	Info
31693	21.565669694	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31694	21.565675701	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29781017 Win=1562624 Len=0 TSval=3611000572 TSecr=3395711037
31695	21.565791918	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31696	21.565915329	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31697	21.565921824	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29783913 Win=1562624 Len=0 TSval=3611000572 TSecr=3395711038
31698	21.566038809	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31699	21.566161312	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31700	21.566167318	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29786809 Win=1562624 Len=0 TSval=3611000572 TSecr=3395711038
31701	21.566284653	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31702	21.566406946	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31703	21.566413023	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29789705 Win=1562624 Len=0 TSval=3611000572 TSecr=3395711038
31704	21.566529728	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31705	21.566653349	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31706	21.566659984	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29792601 Win=1562624 Len=0 TSval=3611000573 TSecr=3395711038
31707	21.566776340	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31708	21.566899472	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31709	21.566905548	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29795497 Win=1562624 Len=0 TSval=3611000573 TSecr=3395711039
31710	21.567022673	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31711	21.567145665	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31712	21.567151741	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29798393 Win=1562624 Len=0 TSval=3611000573 TSecr=3395711039
31713	21.567268726	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31714	21.567391718	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31715	21.567398004	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29801289 Win=1562624 Len=0 TSval=3611000573 TSecr=3395711039
31716	21.567513872	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31717	21.567547256	172.16.1.10	172.16.110.1	FTP-DATA	524	FTP Data: 458 bytes (PASV) (retr files/crab.mp4)
31718	21.567558291	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [ACK] Seq=1 Ack=29803196 Win=1563008 Len=0 TSval=3611000574 TSecr=3395711039
31719	22.023377066	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31720	24.025161479	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31721	24.079017396	Netronix_71:73:ed	Netronix_50:35:0a	ARP	60	Who has 172.16.110.1? Tell 172.16.110.254
31722	24.079026755	Netronix_50:35:0a	Netronix_71:73:ed	ARP	42	172.16.110.1 is at 00:08:54:50:35:0a
31723	26.027244393	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31724	28.029350979	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31725	30.031452671	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31726	32.033553870	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31727	32.164190842	172.16.1.10	172.16.110.1	FTP	89	Response: 226 Transfer complete
31728	32.164221154	172.16.110.1	172.16.1.10	TCP	66	51166 → 21 [ACK] Seq=58 Ack=515 Win=64128 Len=0 TSval=3611011170 TSecr=3395721655
31729	32.164341910	172.16.110.1	172.16.1.10	FTP	72	Request: quit
31730	32.165158712	172.16.1.10	172.16.110.1	FTP	80	Response: 221 Goodbye.
31731	32.165216960	172.16.110.1	172.16.1.10	TCP	66	51166 → 21 [FIN, ACK] Seq=64 Ack=529 Win=64128 Len=0 TSval=3611011171 TSecr=3395721656
31732	32.165237215	172.16.110.1	172.16.1.10	TCP	66	47424 → 38907 [FIN, ACK] Seq=1 Ack=29803196 Win=1563008 Len=0 TSval=3611011171 TSecr=3395711039
31733	32.165441293	172.16.1.10	172.16.110.1	TCP	66	21 → 51166 [FIN, ACK] Seq=529 Ack=64 Win=65280 Len=0 TSval=3395721657 TSecr=3611011170
31734	32.165449185	172.16.110.1	172.16.1.10	TCP	66	51166 → 21 [ACK] Seq=65 Ack=530 Win=64128 Len=0 TSval=3611011171 TSecr=3395721657
31735	32.165472023	172.16.1.10	172.16.110.1	TCP	66	21 → 51166 [ACK] Seq=530 Ack=65 Win=65280 Len=0 TSval=3395721657 TSecr=3611011171
31736	32.368446007	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 47424 → 38907 [FIN, ACK] Seq=1 Ack=29803196 Win=1563008 Len=0 TSval=3611011374 TSecr=3395711039
31737	32.572428578	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 47424 → 38907 [FIN, ACK] Seq=1 Ack=29803196 Win=1563008 Len=0 TSval=3611011578 TSecr=3395711039
31738	32.980436882	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 47424 → 38907 [FIN, ACK] Seq=1 Ack=29803196 Win=1563008 Len=0 TSval=3611011986 TSecr=3395711039
31739	33.812435709	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 47424 → 38907 [FIN, ACK] Seq=1 Ack=29803196 Win=1563008 Len=0 TSval=3611012818 TSecr=3395711039
31740	34.035622099	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001

A.11 Exp6 Step5 Tux2

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
2	2.002160813	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
3	4.004314503	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
4	6.006486001	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
5	8.008215345	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
6	10.010375529	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
7	12.012538158	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
8	13.636483716	172.16.111.1	224.0.0.251	MDNS	160	Standard query 0x0000 PTR _nfs._tcp.local, "QM" question PTR _ftp._tcp.local, "QM" question PTR _webdav._tcp.local, "QM" question PTR _webdavs._tcp.loca
9	14.014699949	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
10	16.016850706	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
11	18.019284383	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
12	20.026003218	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
13	22.024139655	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
14	22.983836548	192.168.88.1	255.255.255.255	MNDP	153	5678 → 5678 Len=111
15	22.983913302	Routerboardc_eb:24:07	CDP/VTP/DTP/PAgP/UDLD	CDP	108	Device ID: MikroTik Port ID: bridge
16	24.025522802	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
17	25.821891200	172.16.111.1	172.16.1.10	TCP	74	38342 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1863385060 TSecr=0 WS=128
18	25.822232367	172.16.1.10	172.16.111.1	TCP	74	21 → 38342 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395302849 TSecr=1863385060 WS=128
19	25.822269173	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1863385060 TSecr=3395302849
20	25.828250529	172.16.1.10	172.16.111.1	FTP	116	Response: 220 ProFTPD Server (Debian) [::ffff:172.16.1.10]
21	25.828261284	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=1 Ack=51 Win=64256 Len=0 TSval=1863385066 TSecr=3395302855
22	25.828363250	172.16.111.1	172.16.1.10	FTP	77	Request: user rcom
23	25.828575842	172.16.1.10	172.16.111.1	TCP	66	21 → 38342 [ACK] Seq=51 Ack=12 Win=65280 Len=0 TSval=3395302856 TSecr=1863385066
24	25.829239320	172.16.1.10	172.16.111.1	FTP	98	Response: 331 Password required for rcom
25	25.829306855	172.16.111.1	172.16.1.10	FTP	77	Request: pass rcom
26	25.873712650	172.16.1.10	172.16.111.1	TCP	66	21 → 38342 [ACK] Seq=83 Ack=23 Win=65280 Len=0 TSval=3395302901 TSecr=1863385067
27	25.977231723	172.16.1.10	172.16.111.1	FTP	113	Response: 230-Welcome, archive user rcom@172.16.1.119 !
28	25.977260637	172.16.1.10	172.16.111.1	FTP	115	Response:
29	25.977318814	172.16.1.10	172.16.111.1	FTP	142	Response:
30	25.977362603	172.16.1.10	172.16.111.1	FTP	157	Response: please report them via e-mail to <root@ftp.netlab.fe.up.pt>.
31	25.977627086	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=23 Ack=346 Win=64128 Len=0 TSval=1863385216 TSecr=3395303004
32	25.977671365	172.16.111.1	172.16.1.10	FTP	74	Request: type A
33	25.977909099	172.16.1.10	172.16.111.1	TCP	66	21 → 38342 [ACK] Seq=346 Ack=31 Win=65280 Len=0 TSval=3395303005 TSecr=1863385216
34	25.978399724	172.16.1.10	172.16.111.1	FTP	85	Response: 200 Type set to A
35	25.978464046	172.16.111.1	172.16.1.10	FTP	72	Request: pasv
36	25.979165377	172.16.1.10	172.16.111.1	FTP	116	Response: 227 Entering Passive Mode (172,16,1,10,156,125).
37	25.979284314	172.16.111.1	172.16.1.10	TCP	74	56824 → 40061 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1863385217 TSecr=0 WS=128
38	25.979548727	172.16.1.10	172.16.111.1	TCP	74	40061 → 56824 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395303007 TSecr=1863385217 WS=128
39	25.979575127	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1863385217 TSecr=3395303007
40	25.979598663	172.16.111.1	172.16.1.10	FTP	81	Request: retr pipe.txt
41	25.980426823	172.16.1.10	172.16.111.1	FTP	131	Response: 150 Opening ASCII mode data connection for pipe.txt (418 bytes)
42	25.980588013	172.16.1.10	172.16.111.1	FTP-DATA	484	FTP Data: 418 bytes (PASV) (retr pipe.txt)
43	25.980598349	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [ACK] Seq=1 Ack=419 Win=64128 Len=0 TSval=1863385218 TSecr=3395303008
44	25.980605473	172.16.1.10	172.16.111.1	TCP	66	40061 → 56824 [FIN, ACK] Seq=419 Ack=1 Win=65280 Len=0 TSval=3395303008 TSecr=1863385217
45	26.024158454	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863385262 TSecr=3395303008
46	26.024178358	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=52 Ack=480 Win=64128 Len=0 TSval=1863385262 TSecr=3395303007
47	26.027652886	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
48	28.029853089	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002

No.	Time	Source	Destination	Protocol	Length	Info
28	25.977260637	172.16.1.10	172.16.111.1	FTP	115	Response:
29	25.977318814	172.16.1.10	172.16.111.1	FTP	142	Response:
30	25.977362603	172.16.1.10	172.16.111.1	FTP	157	Response: please report them via e-mail to <root@ftp.netlab.fe.up.pt>.
31	25.977627086	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=23 Ack=346 Win=64128 Len=0 TSval=1863385216 TSecr=3395303004
32	25.977671365	172.16.111.1	172.16.1.10	FTP	74	Request: type A
33	25.977909099	172.16.1.10	172.16.111.1	TCP	66	21 → 38342 [ACK] Seq=346 Ack=31 Win=65280 Len=0 TSval=3395303005 TSecr=1863385216
34	25.978399724	172.16.1.10	172.16.111.1	FTP	85	Response: 200 Type set to A
35	25.978464046	172.16.111.1	172.16.1.10	FTP	72	Request: pasv
36	25.979165377	172.16.1.10	172.16.111.1	FTP	116	Response: 227 Entering Passive Mode (172,16,1,10,156,125).
37	25.979284314	172.16.111.1	172.16.1.10	TCP	74	56824 → 40061 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1863385217 TSecr=0 WS=128
38	25.979548727	172.16.1.10	172.16.111.1	TCP	74	40061 → 56824 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395303007 TSecr=1863385217 WS=128
39	25.979575127	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1863385217 TSecr=3395303007
40	25.979598663	172.16.111.1	172.16.1.10	FTP	81	Request: retr pipe.txt
41	25.980426823	172.16.1.10	172.16.111.1	FTP	131	Response: 150 Opening ASCII mode data connection for pipe.txt (418 bytes)
42	25.980588013	172.16.1.10	172.16.111.1	FTP-DATA	484	FTP Data: 418 bytes (PASV) (retr pipe.txt)
43	25.980598349	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [ACK] Seq=1 Ack=419 Win=64128 Len=0 TSval=1863385218 TSecr=3395303008
44	25.980605473	172.16.1.10	172.16.111.1	TCP	66	40061 → 56824 [FIN, ACK] Seq=419 Ack=1 Win=65280 Len=0 TSval=3395303008 TSecr=1863385217
45	26.024158454	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863385262 TSecr=3395303008
46	26.024178358	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=52 Ack=480 Win=64128 Len=0 TSval=1863385262 TSecr=3395303007
47	26.027652886	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
48	28.029853089	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
49	30.032012994	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
50	30.814953700	Routerboardc_eb:24:07	KYE_08:d5:98	ARP	60	Who has 172.16.111.1? Tell 172.16.111.254
51	30.814969484	KYE_08:d5:98	Routerboardc_eb:24:07	ARP	42	172.16.111.1 is at 00:c0:df:08:d5:98
52	31.072155022	KYE_08:d5:98	Routerboardc_eb:24:07	ARP	42	Who has 172.16.111.254? Tell 172.16.111.1
53	31.072255731	Routerboardc_eb:24:07	KYE_08:d5:98	ARP	60	172.16.111.254 is at 74:4d:28:eb:24:07
54	32.034142659	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
55	34.036269181	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
56	34.959560563	0.0.0.0	255.255.255.255	MNDP	160	5678 → 5678 Len=118
57	34.959590525	Routerboardc_1c:8d:2c	CDP/VTP/DTP/PAgP/UDLD	CDP	94	Device ID: MikroTik Port ID: bridge111
58	34.959642555	Routerboardc_1c:8d:2c	LLDP_Multicast	LLDP	111	MA/c4:ad:34:1c:8d:2c IN/bridge111 120 SysN=MikroTik SysD=MikroTik RouterOS 6.43.16 (long-term) CRS326-24G-2S+
59	36.038401290	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
60	36.782513465	172.16.1.10	172.16.111.1	FTP	89	Response: 226 Transfer complete
61	36.782548105	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=52 Ack=503 Win=64128 Len=0 TSval=1863396020 TSecr=3395313810
62	36.782657265	172.16.111.1	172.16.1.10	FTP	72	Request: quit
63	36.783207253	172.16.1.10	172.16.111.1	FTP	80	Response: 221 Goodbye.
64	36.783270109	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [FIN, ACK] Seq=58 Ack=517 Win=64128 Len=0 TSval=1863396021 TSecr=3395313811
65	36.783292457	172.16.111.1	172.16.1.10	TCP	66	56824 → 40061 [FIN, ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863396021 TSecr=3395303008
66	36.783459095	172.16.1.10	172.16.111.1	TCP	66	21 → 38342 [FIN, ACK] Seq=517 Ack=58 Win=65280 Len=0 TSval=3395313811 TSecr=1863396021
67	36.783467266	172.16.111.1	172.16.1.10	TCP	66	38342 → 21 [ACK] Seq=59 Ack=518 Win=64128 Len=0 TSval=1863396021 TSecr=3395313811
68	36.783620425	172.16.1.10	172.16.111.1	TCP	66	21 → 38342 [ACK] Seq=518 Ack=59 Win=65280 Len=0 TSval=3395313811 TSecr=1863396021
69	36.984159494	172.16.111.1	172.16.1.10	TCP	66	[TCP Retransmission] 56824 → 40061 [FIN, ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863396222 TSecr=3395303008
70	37.188153536	172.16.111.1	172.16.1.10	TCP	66	[TCP Retransmission] 56824 → 40061 [FIN, ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863396426 TSecr=3395303008
71	37.600167882	172.16.111.1	172.16.1.10	TCP	66	[TCP Retransmission] 56824 → 40061 [FIN, ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863396838 TSecr=3395303008
72	38.040538847	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
73	38.432162847	172.16.111.1	172.16.1.10	TCP	66	[TCP Retransmission] 56824 → 40061 [FIN, ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863397670 TSecr=3395303008
74	40.042688555	Routerboardc_1c:8d:2e	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:07 Cost = 10 Port = 0x8002
75	40.064163164	172.16.111.1	172.16.1.10	TCP	66	[TCP Retransmission] 56824 → 40061 [FIN, ACK] Seq=1 Ack=420 Win=64128 Len=0 TSval=1863399302 TSecr=3395303008

A.12 Exp6 Step5 Tux3

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
2	2.001755536	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
3	4.004240707	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
4	5.807181079	172.16.110.1	172.16.1.10	TCP	74	59086 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3611088922 TSecr=0 WS=128
5	5.807671998	172.16.1.10	172.16.110.1	TCP	74	21 → 59086 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395799412 TSecr=3611088922 WS=128
6	5.807696443	172.16.110.1	172.16.1.10	TCP	66	59086 → 21 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3611088923 TSecr=3395799412
7	5.813944086	172.16.1.10	172.16.110.1	FTP	116	Response: 220 ProFTPD Server (Debian) [::ffff:172.16.1.10]
8	5.813957915	172.16.110.1	172.16.1.10	TCP	66	59086 → 21 [ACK] Seq=1 Ack=51 Win=64256 Len=0 TSval=3611088929 TSecr=3395799418
9	5.814060094	172.16.110.1	172.16.1.10	FTP	77	Request: user rcom
10	5.814308103	172.16.1.10	172.16.110.1	TCP	66	21 → 59086 [ACK] Seq=51 Ack=12 Win=65280 Len=0 TSval=3395799418 TSecr=3611088929
11	5.814992065	172.16.1.10	172.16.110.1	FTP	98	Response: 331 Password required for rcom
12	5.815055761	172.16.110.1	172.16.1.10	FTP	77	Request: pass rcom
13	5.856369172	172.16.1.10	172.16.110.1	TCP	66	21 → 59086 [ACK] Seq=83 Ack=23 Win=65280 Len=0 TSval=3395799461 TSecr=3611088930
14	5.962914775	172.16.1.10	172.16.110.1	FTP	113	Response: 230-Welcome, archive user rcom@172.16.1.119 !
15	5.962946483	172.16.1.10	172.16.110.1	FTP	115	Response:
16	5.962997118	172.16.1.10	172.16.110.1	FTP	142	Response:
17	5.963043494	172.16.1.10	172.16.110.1	FTP	157	Response: please report them via e-mail to <root@ftp.netlab.fe.up.pt>.
18	5.963294855	172.16.110.1	172.16.1.10	TCP	66	59086 → 21 [ACK] Seq=23 Ack=346 Win=64128 Len=0 TSval=3611089078 TSecr=3395799567
19	5.963348493	172.16.110.1	172.16.1.10	FTP	74	Request: type I
20	5.963656008	172.16.1.10	172.16.110.1	TCP	66	21 → 59086 [ACK] Seq=346 Ack=31 Win=65280 Len=0 TSval=3395799568 TSecr=3611089078
21	5.963883832	172.16.1.10	172.16.110.1	FTP	85	Response: 200 Type set to I
22	5.963944595	172.16.110.1	172.16.1.10	FTP	72	Request: pasv
23	5.964616544	172.16.1.10	172.16.110.1	FTP	116	Response: 227 Entering Passive Mode (172,16,1,10,136,171).
24	5.964744914	172.16.110.1	172.16.1.10	TCP	74	54932 → 34987 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3611089080 TSecr=0 WS=128
25	5.965212925	172.16.1.10	172.16.110.1	TCP	74	34987 → 54932 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3395799569 TSecr=3611089080 WS=128
26	5.965227522	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3611089080 TSecr=3395799569
27	5.965250989	172.16.110.1	172.16.1.10	FTP	87	Request: retr files/crab.mp4
28	5.966223189	172.16.1.10	172.16.110.1	FTP	143	Response: 150 Opening BINARY mode data connection for files/crab.mp4 (29803194 bytes)
29	5.966692946	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
30	5.966700768	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 TSval=3611089082 TSecr=3395799570
31	5.966815100	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
32	5.966821665	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=2897 Win=63488 Len=0 TSval=3611089082 TSecr=3395799570
33	5.966939069	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
34	5.966944237	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=4345 Win=62592 Len=0 TSval=3611089082 TSecr=3395799570
35	5.967066321	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
36	5.967071839	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=5793 Win=61568 Len=0 TSval=3611089082 TSecr=3395799570
37	5.967188126	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
38	5.967194132	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=7241 Win=60672 Len=0 TSval=3611089082 TSecr=3395799570
39	5.967312584	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40	5.967318521	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=8689 Win=59648 Len=0 TSval=3611089082 TSecr=3395799571
41	5.967434388	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
42	5.967440115	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=10137 Win=58752 Len=0 TSval=3611089083 TSecr=3395799571
43	5.967557659	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
44	5.967562967	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=11585 Win=57728 Len=0 TSval=3611089083 TSecr=3395799571
45	5.967691197	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
46	5.967696855	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=13033 Win=56832 Len=0 TSval=3611089083 TSecr=3395799571
47	5.967814887	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
48	5.967819916	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=14481 Win=55808 Len=0 TSval=3611089083 TSecr=3395799571

No.	Time	Source	Destination	Protocol	Length	Info
31885	8.508528327	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29788257 Win=3144704 Len=0 TSval=3611091624 TSecr=3395802100
31886	8.508646360	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31887	8.508769631	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31888	8.508774450	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29791153 Win=3144704 Len=0 TSval=3611091624 TSecr=3395802101
31889	8.508892832	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31890	8.509015963	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31891	8.509020922	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29794049 Win=3144704 Len=0 TSval=3611091624 TSecr=3395802101
31892	8.509138815	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31893	8.509261947	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31894	8.509266836	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29796945 Win=3144704 Len=0 TSval=3611091624 TSecr=3395802101
31895	8.509385148	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31896	8.509508349	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31897	8.509513378	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29799841 Win=3144704 Len=0 TSval=3611091625 TSecr=3395802101
31898	8.509631620	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31899	8.509753704	172.16.1.10	172.16.110.1	FTP-DATA	1514	FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
31900	8.509758523	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29802737 Win=3144704 Len=0 TSval=3611091625 TSecr=3395802101
31901	8.509789254	172.16.1.10	172.16.110.1	FTP-DATA	524	FTP Data: 458 bytes (PASV) (retr files/crab.mp4)
31902	8.551306743	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611091666 TSecr=3395802102
31903	10.010669102	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31904	10.849837068	Netronix_71:73:ed	Netronix_50:35:0a	ARP	60	Who has 172.16.110.1? Tell 172.16.110.254
31905	10.849852084	Netronix_50:35:0a	Netronix_71:73:ed	ARP	42	172.16.110.1 is at 00:08:54:50:35:0a
31906	12.012166902	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31907	14.014829732	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31908	16.016925721	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31909	18.019014932	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31910	19.139427983	172.16.1.10	172.16.110.1	FTP	89	Response: 226 Transfer complete
31911	19.139460878	172.16.110.1	172.16.1.10	TCP	66	59086 → 21 [ACK] Seq=58 Ack=515 Win=64128 Len=0 TSval=3611102255 TSecr=3395812744
31912	19.139580448	172.16.110.1	172.16.1.10	FTP	72	Request: quit
31913	19.140207070	172.16.1.10	172.16.110.1	FTP	80	Response: 221 Goodbye.
31914	19.140265388	172.16.110.1	172.16.1.10	TCP	66	59086 → 21 [FIN, ACK] Seq=64 Ack=529 Win=64128 Len=0 TSval=3611102255 TSecr=3395812745
31915	19.140289274	172.16.110.1	172.16.1.10	TCP	66	54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611102255 TSecr=3395802102
31916	19.140514444	172.16.1.10	172.16.110.1	TCP	66	21 → 59086 [FIN, ACK] Seq=529 Ack=64 Win=65280 Len=0 TSval=3395812745 TSecr=3611102255
31917	19.140522406	172.16.110.1	172.16.1.10	TCP	66	59086 → 21 [ACK] Seq=65 Ack=530 Win=64128 Len=0 TSval=3611102256 TSecr=3395812745
31918	19.140530368	172.16.1.10	172.16.110.1	TCP	66	21 → 59086 [ACK] Seq=530 Ack=65 Win=65280 Len=0 TSval=3395812745 TSecr=3611102255
31919	19.343295645	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611102458 TSecr=3395802102
31920	19.547299429	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611102662 TSecr=3395802102
31921	19.975311483	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611103090 TSecr=3395802102
31922	20.021134520	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31923	20.807319870	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611103922 TSecr=3395802102
31924	22.023240974	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31925	22.439312634	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611105554 TSecr=3395802102
31926	24.025353291	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31927	25.863307014	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611108978 TSecr=3395802102
31928	26.027455756	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31929	28.029548789	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31930	30.031654949	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31931	32.033759010	Routerboardc_1c:8d:37	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8d:37 Cost = 0 Port = 0x8001
31932	32.519316651	172.16.110.1	172.16.1.10	TCP	66	[TCP Retransmission] 54932 → 34987 [FIN, ACK] Seq=1 Ack=29803196 Win=3145728 Len=0 TSval=3611115634 TSecr=3395802102

A.13 Exp2 Step10 Tux2

38	68.065262197	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
39	70.067474621	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
40	72.069685997	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
41	74.071894999	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
42	76.074106794	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
43	78.076317681	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
44	80.078526683	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
45	82.080735475	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
46	82.293047354	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=1/256, ttl=64 (no response found!)
47	83.297058437	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=2/512, ttl=64 (no response found!)
48	84.082953556	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
49	84.321038480	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=3/768, ttl=64 (no response found!)
50	85.345052955	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=4/1024, ttl=64 (no response found!)
51	86.085164792	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
52	86.369060027	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=5/1280, ttl=64 (no response found!)
53	87.393057600	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=6/1536, ttl=64 (no response found!)
54	88.087379311	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
55	88.417058805	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=7/1792, ttl=64 (no response found!)
56	89.441057217	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=8/2048, ttl=64 (no response found!)
57	90.089592782	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
58	90.469058765	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=9/2304, ttl=64 (no response found!)
59	91.489059976	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=10/2560, ttl=64 (no response found!)
60	92.091801714	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
61	92.513058806	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=11/2816, ttl=64 (no response found!)
62	93.537054214	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=12/3072, ttl=64 (no response found!)
63	94.094020632	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
64	94.561061635	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=13/3328, ttl=64 (no response found!)
65	95.585059418	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=14/3584, ttl=64 (no response found!)
66	96.096228586	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
67	96.609059017	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=15/3840, ttl=64 (no response found!)
68	97.633059523	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=16/4096, ttl=64 (no response found!)
69	98.098442546	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
70	98.657057725	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=17/4352, ttl=64 (no response found!)
71	99.681066683	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=18/4608, ttl=64 (no response found!)
72	100.100655039	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
73	100.705060973	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=19/4864, ttl=64 (no response found!)
74	101.729053727	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=20/5120, ttl=64 (no response found!)
75	102.102859710	Routerboardc_2b:fa:…	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8001
76	102.753057796	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x17cf, seq=21/5376, ttl=64 (no response found!)

A.14 Exp2 Step8 Tux3

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	0.647227178	0.0.0.0	255.255.255.255	MNDP	159	5678 → 5678 Len=117
3	0.647266220	Routerboardc_2b:fa:06	CDP/VTP/DTP/PAgP/UDLD	CDP	93	Device ID: MikroTik Port ID: bridge20
4	0.647316087	Routerboardc_2b:fa:06	LLDP_Multicast	LLDP	110	MA/c4:ad:34:2b:fa:06 IN/bridge20 120 SysN=MikroTik SysD=MikroTik RouterOS 6.43.
5	2.002207217	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
6	4.004420459	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
7	6.006624917	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
8	8.008838543	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
9	10.011040522	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
10	12.013253763	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
11	14.015458221	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
12	16.017659065	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
13	18.019869564	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
14	20.022073235	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
15	20.846638161	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=1/256, ttl=64 (no response found!)
16	20.846825827	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=1/256, ttl=64
17	21.858467024	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=2/512, ttl=64 (no response found!)
18	21.858612715	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=2/512, ttl=64
19	22.024279577	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
20	22.882462531	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=3/768, ttl=64 (no response found!)
21	22.882613669	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=3/768, ttl=64
22	23.906466004	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=4/1024, ttl=64 (no response found!)
23	23.906646546	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=4/1024, ttl=64
24	24.026483841	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
25	24.930464872	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=5/1280, ttl=64 (no response found!)
26	24.930613427	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=5/1280, ttl=64
27	25.909944484	Netronix_c8:7c:55	KYE_13:20:10	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
28	25.909964668	KYE_13:20:10	Netronix_c8:7c:55	ARP	42	172.16.20.1 is at 00:c0:df:13:20:10
29	25.954453967	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=6/1536, ttl=64 (no response found!)
30	25.954592324	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=6/1536, ttl=64
31	26.028679392	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
32	26.978462412	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=7/1792, ttl=64 (no response found!)
33	26.978610547	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=7/1792, ttl=64
34	28.002463668	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=8/2048, ttl=64 (no response found!)
35	28.002609847	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=8/2048, ttl=64
36	28.030918610	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
37	29.026462414	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=9/2304, ttl=64 (no response found!)
38	29.026632898	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=9/2304, ttl=64
39	30.033087444	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
40	30.050461094	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=10/2560, ttl=64 (no response found!)
41	30.050602105	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=10/2560, ttl=64
42	31.074465226	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=11/2816, ttl=64 (no response found!)
43	31.074604282	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=11/2816, ttl=64
44	32.035291845	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
45	32.098461820	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=12/3072, ttl=64 (no response found!)
46	32.098602482	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=12/3072, ttl=64
47	33.122486704	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=13/3328, ttl=64 (no response found!)
48	33.122630439	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=13/3328, ttl=64

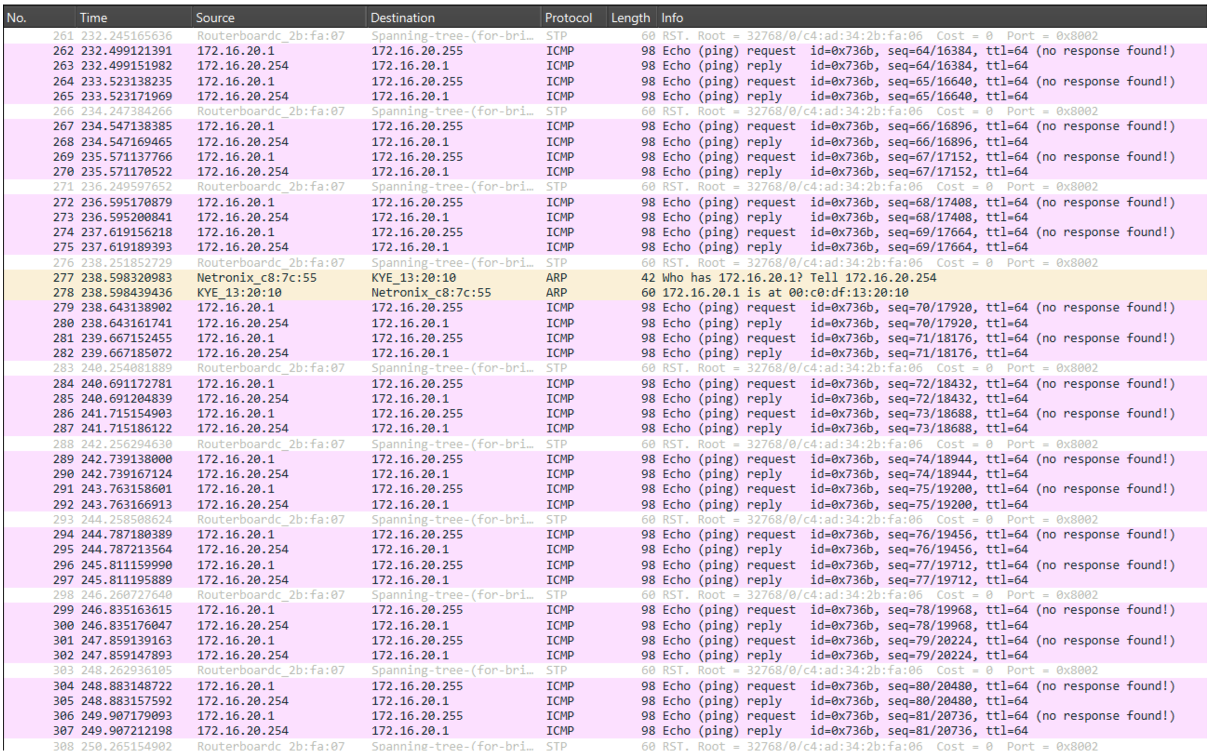
A.15 Exp2 Step10 Tux3

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=195/49920, ttl=64 (no response found!)
2	0.000168460	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=195/49920, ttl=64
3	0.740637620	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
4	1.023995165	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=196/50176, ttl=64 (no response found!)
5	1.024128703	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=196/50176, ttl=64
6	2.047996968	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=197/50432, ttl=64 (no response found!)
7	2.048133090	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=197/50432, ttl=64
8	2.742832580	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
9	3.071995002	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=198/50688, ttl=64 (no response found!)
10	3.072123791	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=198/50688, ttl=64
11	4.095994715	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=199/50944, ttl=64 (no response found!)
12	4.096134260	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=199/50944, ttl=64
13	4.745031671	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
14	5.119992476	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=200/51200, ttl=64 (no response found!)
15	5.120153602	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=200/51200, ttl=64
16	6.144007631	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=201/51456, ttl=64 (no response found!)
17	6.144142356	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=201/51456, ttl=64
18	6.747227421	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
19	7.167993943	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=202/51712, ttl=64 (no response found!)
20	7.168134606	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=202/51712, ttl=64
21	8.191999256	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=203/51968, ttl=64 (no response found!)
22	8.192139569	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=203/51968, ttl=64
23	8.749429469	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
24	9.215994584	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=204/52224, ttl=64 (no response found!)
25	9.216161787	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=204/52224, ttl=64
26	10.239999903	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=205/52480, ttl=64 (no response found!)
27	10.240137213	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=205/52480, ttl=64
28	10.751634880	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
29	11.264008158	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=206/52736, ttl=64 (no response found!)
30	11.264143023	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=206/52736, ttl=64
31	12.287996790	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=207/52992, ttl=64 (no response found!)
32	12.288134100	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=207/52992, ttl=64
33	12.526762276	fe80::2e0:7dff:fec8:7c…	ff02::fb	MDNS	180	Standard query 0x0000 PTR _ftp._tcp.local, "QM" question PTR _nfs._tcp.local, "QM" question PTR _afpovertcp._tcp.local, "QM" question PTR _smb._tcp.loca
34	12.753831921	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
35	13.312001908	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=208/53248, ttl=64 (no response found!)
36	13.312142780	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=208/53248, ttl=64
37	14.335998997	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=209/53504, ttl=64 (no response found!)
38	14.336166828	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=209/53504, ttl=64
39	14.756032116	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
40	15.359998045	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=210/53760, ttl=64 (no response found!)
41	15.360135494	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=210/53760, ttl=64
42	16.383998981	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=211/54016, ttl=64 (no response found!)
43	16.384140271	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=211/54016, ttl=64
44	16.758230786	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
45	17.407998593	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=212/54272, ttl=64 (no response found!)
46	17.408138068	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=212/54272, ttl=64
47	18.436001352	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=213/54528, ttl=64 (no response found!)
48	18.436138802	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=213/54528, ttl=64
49	18.760443086	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001

No.	Time	Source	Destination	Protocol	Length	Info
338	132.095995448	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=324/17409, ttl=64 (no response found!)
339	132.096154549	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=324/17409, ttl=64
340	132.876789648	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
341	133.119991792	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=325/17665, ttl=64 (no response found!)
342	133.120161089	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=325/17665, ttl=64
343	134.143998055	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=326/17921, ttl=64 (no response found!)
344	134.144172032	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=326/17921, ttl=64
345	134.878997704	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
346	135.168010188	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=327/18177, ttl=64 (no response found!)
347	135.168188844	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=327/18177, ttl=64
348	136.191997458	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=328/18433, ttl=64 (no response found!)
349	136.192161099	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=328/18433, ttl=64
350	136.881206397	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
351	137.216012738	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=329/18689, ttl=64 (no response found!)
352	137.216192023	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=329/18689, ttl=64
353	138.239997988	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=330/18945, ttl=64 (no response found!)
354	138.240156670	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=330/18945, ttl=64
355	138.883415308	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
356	139.264005590	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=331/19201, ttl=64 (no response found!)
357	139.264181732	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=331/19201, ttl=64
358	140.292012219	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=332/19457, ttl=64 (no response found!)
359	140.292156443	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=332/19457, ttl=64
360	140.885623390	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
361	141.162279395	0.0.0.0	255.255.255.255	MNDP	159	5678 → 5678 Len=117
362	141.162311732	Routerboardc_2b:fa:06	CDP/VTP/DTP/PAgP/UDLD	CDP	93	Device ID: MikroTik Port ID: bridge20
363	141.162362088	Routerboardc_2b:fa:06	LLDP_Multicast	LLDP	110	MA/c4:ad:34:2b:fa:06 IN/bridge20 120 SysN=MikroTik SysD=MikroTik RouterOS 6.43.16 (long-t
364	141.311995308	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=333/19713, ttl=64 (no response found!)
365	141.312133456	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=333/19713, ttl=64
366	142.335999215	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=334/19969, ttl=64 (no response found!)
367	142.336139738	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=334/19969, ttl=64
368	142.887834274	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
369	143.359999283	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=335/20225, ttl=64 (no response found!)
370	143.360143647	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=335/20225, ttl=64
371	144.383999214	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=336/20481, ttl=64 (no response found!)
372	144.384167883	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=336/20481, ttl=64
373	144.889802535	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
374	145.407999147	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=337/20737, ttl=64 (no response found!)
375	145.408140438	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=337/20737, ttl=64
376	146.432001317	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=338/20993, ttl=64 (no response found!)
377	146.432175713	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=338/20993, ttl=64
378	146.892012808	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
379	147.455998530	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=339/21249, ttl=64 (no response found!)
380	147.456142405	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=339/21249, ttl=64
381	148.480008388	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=340/21505, ttl=64 (no response found!)
382	148.480147513	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=340/21505, ttl=64
383	148.894218760	Routerboardc_2b:fa:06	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
384	149.503997504	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=341/21761, ttl=64 (no response found!)
385	149.504173716	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=341/21761, ttl=64

A.16 Exp2 Step8 Tux4

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
2	2.002214912	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
3	4.004437365	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
4	6.006652135	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
5	8.008856916	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
6	10.011069240	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
7	12.013285544	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
8	14.015505827	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
9	16.017713050	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
10	18.019923484	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
11	20.022132031	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
12	22.024347911	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
13	24.026560927	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
14	26.028776176	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
15	27.797890588	0.0.0.0	255.255.255.255	MNDP	159	5678 → 5678 Len=117
16	27.797906722	Routerboardc_2b:fa:06	CDP/VTP/DTP/PAgP/UDLD	CDP	93	Device ID: MikroTik Port ID: bridge20
17	27.797964761	Routerboardc_2b:fa:06	LLDP_Multicast	LLDP	110	MA/c4:ad:34:2b:fa:06 IN/bridge20 120 SysN=MikroTik SysD=MikroTik RouterOS 6.43.16 (long-term) CRS326-24G-2S+
18	28.030980111	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
19	30.033203251	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
20	32.035417310	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
21	34.037635071	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
22	36.039851503	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
23	38.042148253	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
24	40.044504997	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
25	42.046855803	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
26	44.049208634	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
27	46.041483023	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
28	48.043690580	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
29	50.045910778	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
30	52.048121197	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
31	54.050334968	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
32	56.052555932	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
33	58.054763066	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
34	60.056979558	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
35	62.059194862	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
36	62.868230033	172.16.20.1	224.0.0.251	MDNS	160	Standard query 0x0000 PTR _nfs._tcp.local, "QM" question PTR _ftp._tcp.local, "QM" question PTR _webdav._tcp.local, "QM" question PTR _webdavs._tcp.local
37	64.061409885	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
38	66.063625048	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
39	68.065222315	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
40	70.067440549	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
41	72.069656547	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
42	74.071871497	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
43	76.074087563	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
44	78.076296924	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
45	80.078508519	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
46	82.080721580	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
47	84.082932195	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
48	86.029062026	172.16.20.254	224.0.0.251	MDNS	160	Standard query 0x0000 PTR _nfs._tcp.local, "QM" question PTR _ftp._tcp.local, "QM" question PTR _webdav._tcp.local, "QM" question PTR _webdavs._tcp.local



No.	Time	Source	Destination	Protocol	Length	Info
261	232.245165636	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
262	232.499121391	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=64/16384, ttl=64 (no response found!)
263	232.499151982	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=64/16384, ttl=64
264	233.523138235	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=65/16640, ttl=64 (no response found!)
265	233.523171969	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=65/16640, ttl=64
266	234.247384266	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
267	234.547138385	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=66/16896, ttl=64 (no response found!)
268	234.547169465	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=66/16896, ttl=64
269	235.571137766	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=67/17152, ttl=64 (no response found!)
270	235.571170522	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=67/17152, ttl=64
271	236.249597652	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
272	236.595170879	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=68/17408, ttl=64 (no response found!)
273	236.595200841	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=68/17408, ttl=64
274	237.619156218	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=69/17664, ttl=64 (no response found!)
275	237.619189393	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=69/17664, ttl=64
276	238.251852729	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
277	238.598320983	Netronix_c8:7c:55	KYE_13:20:10	ARP	42	Who has 172.16.20.1? Tell 172.16.20.254
278	238.598439436	KYE_13:20:10	Netronix_c8:7c:55	ARP	60	172.16.20.1 is at 00:c0:df:13:20:10
279	238.643138902	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=70/17920, ttl=64 (no response found!)
280	238.643161741	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=70/17920, ttl=64
281	239.667152455	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=71/18176, ttl=64 (no response found!)
282	239.667185072	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=71/18176, ttl=64
283	240.254081889	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
284	240.691172781	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=72/18432, ttl=64 (no response found!)
285	240.691204839	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=72/18432, ttl=64
286	241.715154903	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=73/18688, ttl=64 (no response found!)
287	241.715186122	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=73/18688, ttl=64
288	242.256294630	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
289	242.739138000	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=74/18944, ttl=64 (no response found!)
290	242.739167124	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=74/18944, ttl=64
291	243.763158601	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=75/19200, ttl=64 (no response found!)
292	243.763166913	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=75/19200, ttl=64
293	244.258508624	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
294	244.787180389	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=76/19456, ttl=64 (no response found!)
295	244.787213564	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=76/19456, ttl=64
296	245.811159990	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=77/19712, ttl=64 (no response found!)
297	245.811195889	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=77/19712, ttl=64
298	246.260727640	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
299	246.835163615	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=78/19968, ttl=64 (no response found!)
300	246.835176047	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=78/19968, ttl=64
301	247.859139163	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=79/20224, ttl=64 (no response found!)
302	247.859147893	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=79/20224, ttl=64
303	248.262936105	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002
304	248.883148722	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=80/20480, ttl=64 (no response found!)
305	248.883157592	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=80/20480, ttl=64
306	249.907179093	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x736b, seq=81/20736, ttl=64 (no response found!)
307	249.907212198	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x736b, seq=81/20736, ttl=64
308	250.265154902	Routerboardc_2b:fa:07	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8002

A.17 Exp2 Step8-10 Tux4

A.18 Exp3 Step7 Tux3

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
2	2.002372814	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
3	4.004112097	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
4	6.006492098	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
5	8.008874471	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
6	10.011252091	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
7	11.984720368	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=1/256, ttl=64 (reply in 8)
8	11.984888825	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=1/256, ttl=64 (request in 7)
9	12.012956579	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
10	12.987882083	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=2/512, ttl=64 (reply in 11)
11	12.988020508	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=2/512, ttl=64 (request in 10)
12	14.011878524	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=3/768, ttl=64 (reply in 13)
13	14.012021140	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=3/768, ttl=64 (request in 12)
14	14.015318339	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
15	15.035876221	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=4/1024, ttl=64 (reply in 16)
16	15.036045167	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=4/1024, ttl=64 (request in 15)
17	16.017687639	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
18	16.059874966	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=5/1280, ttl=64 (reply in 19)
19	16.060003334	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=5/1280, ttl=64 (request in 18)
20	16.987843758	KYE_25:40:66	3Com_a1:35:69	ARP	42	Who has 172.16.60.254? Tell 172.16.60.1
21	16.987965072	3Com_a1:35:69	KYE_25:40:66	ARP	60	172.16.60.254 is at 00:01:02:a1:35:69
22	17.037335071	3Com_a1:35:69	KYE_25:40:66	ARP	60	Who has 172.16.60.1? Tell 172.16.60.254
23	17.037342544	KYE_25:40:66	3Com_a1:35:69	ARP	42	172.16.60.1 is at 00:c0:df:25:40:66
24	18.020061894	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
25	20.022437543	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
26	22.024200682	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
27	23.088257920	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=1/256, ttl=64 (reply in 28)
28	23.088416180	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=1/256, ttl=64 (request in 27)
29	24.026586312	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
30	24.091872196	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=2/512, ttl=64 (reply in 31)
31	24.091996723	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=2/512, ttl=64 (request in 30)
32	25.115880012	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=3/768, ttl=64 (reply in 33)
33	25.116046234	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=3/768, ttl=64 (request in 32)
34	26.028966282	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
35	26.139871276	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=4/1024, ttl=64 (reply in 36)
36	26.140001669	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=4/1024, ttl=64 (request in 35)
37	27.163870081	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=5/1280, ttl=64 (reply in 38)
38	27.164003827	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=5/1280, ttl=64 (request in 37)
39	28.031348413	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
40	30.033070264	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
41	32.035436116	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
42	32.736464381	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=1/256, ttl=64 (reply in 43)
43	32.736773358	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=1/256, ttl=63 (request in 42)
44	33.755873711	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=2/512, ttl=64 (reply in 45)
45	33.756125278	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=2/512, ttl=63 (request in 44)
46	34.037813070	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
47	34.779879494	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=3/768, ttl=64 (reply in 48)
48	34.780161023	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=3/768, ttl=63 (request in 47)
49	35.803871797	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=4/1024, ttl=64 (reply in 50)
50	35.804146063	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=4/1024, ttl=63 (request in 49)
51	36.040181291	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
52	36.827875693	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=5/1280, ttl=64 (reply in 53)
53	36.828132848	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=5/1280, ttl=63 (request in 52)
54	37.034799112	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
55	37.034837385	Routerboardc_1c:8b:e5	CDP/VTP/DTP/PAgP/UDLD	CDP	116	Device ID: MikroTik Port ID: bridge60/ether3
56	37.034925594	Routerboardc_1c:8b:e5	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:8b:e3 IN/bridge60/ether3 120 SysN=MikroTik SysD=MikroTi
57	37.851873791	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=6/1536, ttl=64 (reply in 58)
58	37.852118585	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=6/1536, ttl=63 (request in 57)
59	38.042542385	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
60	40.044307452	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
61	42.046675035	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001

A.19 Exp3 Step10 Tux4 eth1

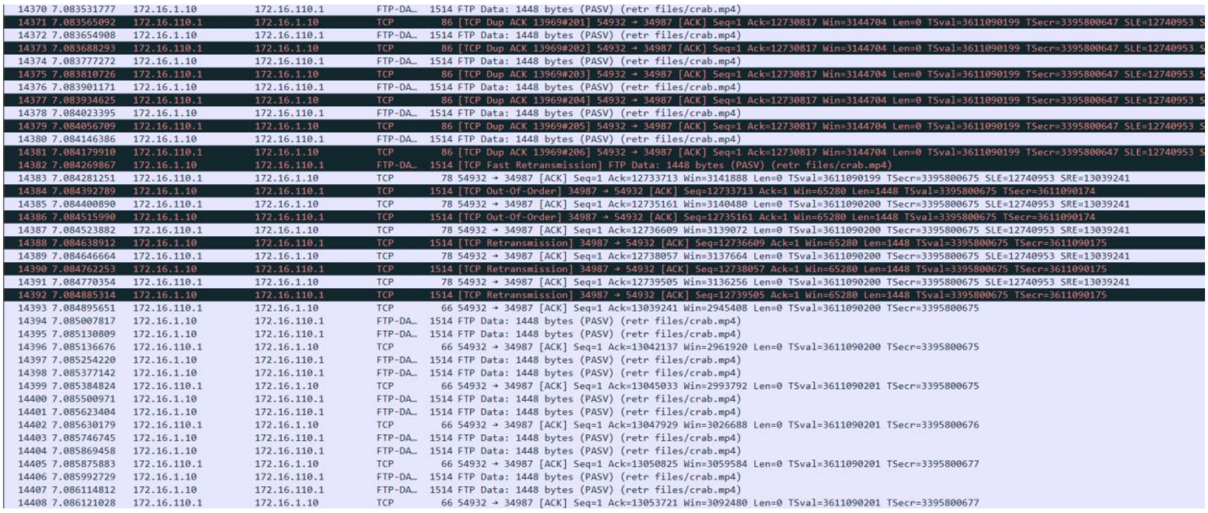
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
2	2.002372814	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
3	4.004112097	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
4	6.006492098	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
5	8.008874471	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
6	10.011252091	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
7	11.984720368	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=1/256, ttl=64 (reply in 8)
8	11.984888825	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=1/256, ttl=64 (request in 7)
9	12.012956579	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
10	12.987882083	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=2/512, ttl=64 (reply in 11)
11	12.988020508	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=2/512, ttl=64 (request in 10)
12	14.011878524	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=3/768, ttl=64 (reply in 13)
13	14.012021140	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=3/768, ttl=64 (request in 12)
14	14.015318339	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
15	15.035876221	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=4/1024, ttl=64 (reply in 16)
16	15.036045167	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=4/1024, ttl=64 (request in 15)
17	16.017687639	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
18	16.059874966	172.16.60.1	172.16.60.254	ICMP	98	Echo (ping) request id=0x51c4, seq=5/1280, ttl=64 (reply in 19)
19	16.060003334	172.16.60.254	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51c4, seq=5/1280, ttl=64 (request in 18)
20	16.987843758	KYE_25:40:66	3Com_a1:35:69	ARP	42	Who has 172.16.60.254? Tell 172.16.60.1
21	16.987965072	3Com_a1:35:69	KYE_25:40:66	ARP	60	172.16.60.254 is at 00:01:02:a1:35:69
22	17.037335071	3Com_a1:35:69	KYE_25:40:66	ARP	60	Who has 172.16.60.1? Tell 172.16.60.254
23	17.037342544	KYE_25:40:66	3Com_a1:35:69	ARP	42	172.16.60.1 is at 00:c0:df:25:40:66
24	18.020061894	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
25	20.022437543	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
26	22.024200682	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
27	23.088257920	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=1/256, ttl=64 (reply in 28)
28	23.088416180	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=1/256, ttl=64 (request in 27)
29	24.026586312	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
30	24.091872196	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=2/512, ttl=64 (reply in 31)
31	24.091996723	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=2/512, ttl=64 (request in 30)
32	25.115880012	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=3/768, ttl=64 (reply in 33)
33	25.116046234	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=3/768, ttl=64 (request in 32)
34	26.028966282	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
35	26.139871276	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=4/1024, ttl=64 (reply in 36)
36	26.140001669	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=4/1024, ttl=64 (request in 35)
37	27.163870081	172.16.60.1	172.16.61.253	ICMP	98	Echo (ping) request id=0x51cb, seq=5/1280, ttl=64 (reply in 38)
38	27.164003827	172.16.61.253	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51cb, seq=5/1280, ttl=64 (request in 37)
39	28.031348413	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
40	30.033070264	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
41	32.035436116	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
42	32.736464381	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=1/256, ttl=64 (reply in 43)
43	32.736773358	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=1/256, ttl=63 (request in 42)
44	33.755873711	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=2/512, ttl=64 (reply in 45)
45	33.756125278	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=2/512, ttl=63 (request in 44)
46	34.037813070	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
47	34.779879494	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=3/768, ttl=64 (reply in 48)
48	34.780161023	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=3/768, ttl=63 (request in 47)
49	35.803871797	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=4/1024, ttl=64 (reply in 50)
50	35.804146063	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=4/1024, ttl=63 (request in 49)
51	36.040181291	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
52	36.827875693	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=5/1280, ttl=64 (reply in 53)
53	36.828132848	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=5/1280, ttl=63 (request in 52)
54	37.034799112	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
55	37.034837385	Routerboardc_1c:8b:e5	CDP/VTP/DTP/PAgP/UDLD	CDP	116	Device ID: MikroTik Port ID: bridge60/ether3
56	37.034925594	Routerboardc_1c:8b:e5	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:8b:e3 IN/bridge60/ether3 120 SysN=MikroTik SysD=MikroTik Rou
57	37.851873791	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x51d2, seq=6/1536, ttl=64 (reply in 58)
58	37.852118585	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x51d2, seq=6/1536, ttl=63 (request in 57)
59	38.042542385	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
60	40.044307452	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001
61	42.046675035	Routerboardc_1c:8b:e5	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e5 Cost = 0 Port = 0x0001

A.20 Exp3 Step10 Tux4 eth2

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
2	2.002375822	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
3	4.004410251	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
4	5.743075925	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
5	5.743144581	Routerboardc_1c:8b:e4	CDP/VTP/DTP/PAgP/UDLD	CDP	116	Device ID: MikroTik Port ID: bridge61/ether2
6	5.743234329	Routerboardc_1c:8b:e4	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:8b:e3 IN/bridge61/ether2 120 SysN=MikroTik SysD=MikroTik RouterOS
7	6.006772590	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
8	8.008856115	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
9	10.011234864	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
10	12.013282836	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
11	14.015659418	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
12	16.018037744	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
13	18.020107013	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
14	20.022476466	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
15	22.024500893	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
16	24.026880680	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
17	26.028967195	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
18	28.031343906	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
19	30.033362391	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
20	32.035742591	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
21	34.038122300	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
22	36.040201824	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
23	38.042578038	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
24	40.044591208	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
25	42.046968118	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
26	44.049062303	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
27	46.051432854	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
28	48.053463409	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
29	50.055834587	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
30	52.058217147	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
31	54.060304061	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
32	56.062675933	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
33	58.064706132	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
34	60.067079049	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
35	62.069175945	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
36	64.071550395	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
37	65.747440489	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
38	65.747471360	Routerboardc_1c:8b:e4	CDP/VTP/DTP/PAgP/UDLD	CDP	116	Device ID: MikroTik Port ID: bridge61/ether2
39	65.747561736	Routerboardc_1c:8b:e4	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:8b:e3 IN/bridge61/ether2 120 SysN=MikroTik SysD=MikroTik RouterOS
40	66.073548391	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
41	68.075934014	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
42	70.078305666	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
43	72.080408283	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
44	74.082785450	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
45	76.084812424	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
46	78.087190986	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
47	80.089265102	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
48	82.091637165	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
49	84.093667976	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002

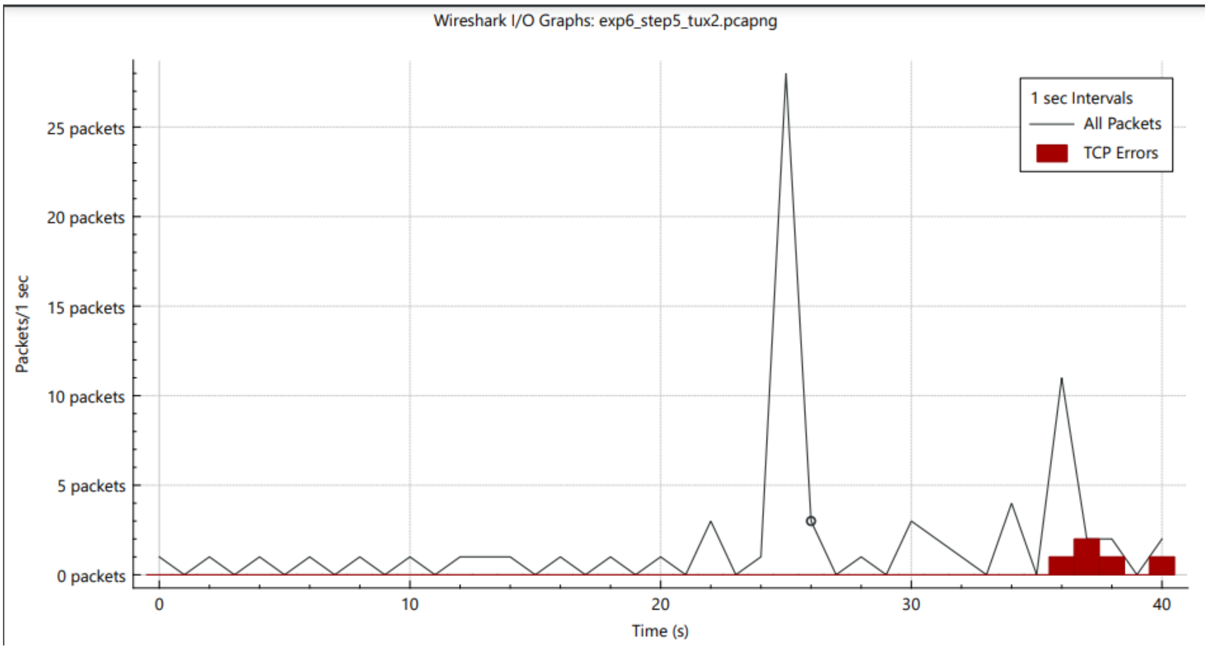
No.	Time	Source	Destination	Protocol	Length	Info
37	65.747440489	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
38	65.747471360	Routerboardc_1c:8b:e4	CDP/VTP/DTP/PAgP/UDLD	CDP	116	Device ID: MikroTik Port ID: bridge61/ether2
39	65.747561736	Routerboardc_1c:8b:e4	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:8b:e3 IN/bridge61/ether2 120 SysN=MikroTik SysD=MikroTik Router
40	66.073548391	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
41	68.075934014	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
42	70.078305666	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
43	72.080408283	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
44	74.082785450	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
45	76.084812424	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
46	78.087190986	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
47	80.089265102	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
48	82.091637165	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
49	84.093667976	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
50	86.096042272	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
51	88.098418732	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
52	90.100506460	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
53	92.102882568	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
54	94.104903664	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
55	96.107286823	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
56	98.109369169	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
57	100.111740941	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
58	102.113772019	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
59	104.116148120	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
60	105.999378355	KYE_04:20:8c	Broadcast	ARP	42	Who has 172.16.61.1? Tell 172.16.61.253
61	105.999524396	Netronix_b5:8c:8e	KYE_04:20:8c	ARP	60	172.16.61.1 is at 00:e0:7d:b5:8c:8e
62	105.999540390	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=1/256, ttl=63 (reply in 63)
63	105.999659821	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=1/256, ttl=64 (request in 62)
64	106.118496422	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
65	107.030600201	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=2/512, ttl=63 (reply in 66)
66	107.030716629	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=2/512, ttl=64 (request in 65)
67	108.054607837	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=3/768, ttl=63 (reply in 68)
68	108.054722519	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=3/768, ttl=64 (request in 67)
69	108.120577643	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
70	109.078612819	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=4/1024, ttl=63 (reply in 71)
71	109.078726733	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=4/1024, ttl=64 (request in 70)
72	110.102615286	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=5/1280, ttl=63 (reply in 73)
73	110.102751270	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=5/1280, ttl=64 (request in 72)
74	110.122921263	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
75	111.126573542	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=6/1536, ttl=63 (reply in 76)
76	111.126685570	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=6/1536, ttl=64 (request in 75)
77	111.151214946	Netronix_b5:8c:8e	KYE_04:20:8c	ARP	60	Who has 172.16.61.253? Tell 172.16.61.1
78	111.151222559	KYE_04:20:8c	Netronix_b5:8c:8e	ARP	42	172.16.61.253 is at 00:c0:df:04:20:8c
79	112.125004368	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
80	112.150607787	172.16.60.1	172.16.61.1	ICMP	98	Echo (ping) request id=0x5300, seq=7/1792, ttl=63 (reply in 81)
81	112.150718628	172.16.61.1	172.16.60.1	ICMP	98	Echo (ping) reply id=0x5300, seq=7/1792, ttl=64 (request in 80)
82	114.127384793	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
83	116.129467825	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002
84	118.131843149	Routerboardc_1c:8b:e4	Spanning-tree-(for-bri…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:8b:e3 Cost = 0 Port = 0x0002

A.21 Data phase of application

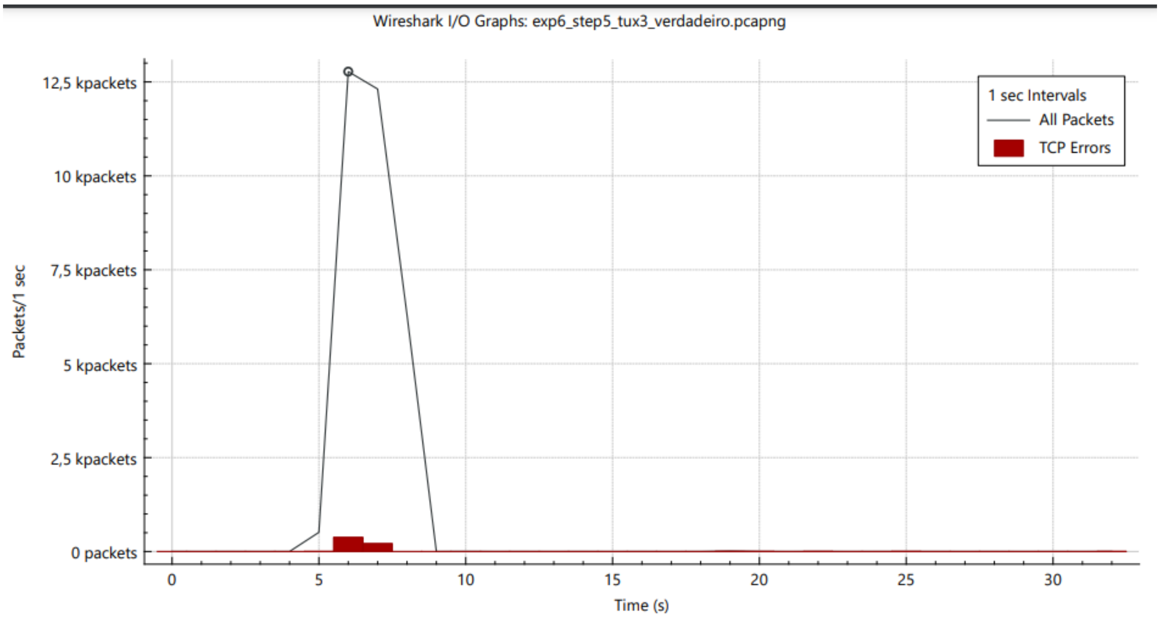


14370	7.083531777	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14371	7.083565092	172.16.110.1	172.16.1.10	TCP	86 [TCP Dup ACK 13969#201] 54932 → 34987 [ACK] Seq=1 Ack=12730817 Win=3144704 Len=0 TSval=3611090199 TSecr=3395800647 SLE=12740953 S
14372	7.083654908	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14373	7.083688293	172.16.110.1	172.16.1.10	TCP	86 [TCP Dup ACK 13969#202] 54932 → 34987 [ACK] Seq=1 Ack=12730817 Win=3144704 Len=0 TSval=3611090199 TSecr=3395800647 SLE=12740953 S
14374	7.083777272	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14375	7.083810726	172.16.110.1	172.16.1.10	TCP	86 [TCP Dup ACK 13969#203] 54932 → 34987 [ACK] Seq=1 Ack=12730817 Win=3144704 Len=0 TSval=3611090199 TSecr=3395800647 SLE=12740953 S
14376	7.083901171	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14377	7.083934625	172.16.110.1	172.16.1.10	TCP	86 [TCP Dup ACK 13969#204] 54932 → 34987 [ACK] Seq=1 Ack=12730817 Win=3144704 Len=0 TSval=3611090199 TSecr=3395800647 SLE=12740953 S
14378	7.084023395	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14379	7.084056709	172.16.110.1	172.16.1.10	TCP	86 [TCP Dup ACK 13969#205] 54932 → 34987 [ACK] Seq=1 Ack=12730817 Win=3144704 Len=0 TSval=3611090199 TSecr=3395800647 SLE=12740953 S
14380	7.084146386	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14381	7.084179910	172.16.110.1	172.16.1.10	TCP	86 [TCP Dup ACK 13969#206] 54932 → 34987 [ACK] Seq=1 Ack=12730817 Win=3144704 Len=0 TSval=3611090199 TSecr=3395800647 SLE=12740953 S
14382	7.084269867	172.16.1.10	172.16.110.1	FTP-DA…	1514 [TCP Fast Retransmission] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14383	7.084281251	172.16.110.1	172.16.1.10	TCP	78 54932 → 34987 [ACK] Seq=1 Ack=12733713 Win=3141888 Len=0 TSval=3611090199 TSecr=3395800675 SLE=12740953 SRE=13039241
14384	7.084392789	172.16.1.10	172.16.110.1	TCP	1514 [TCP Out-Of-Order] 34987 → 54932 [ACK] Seq=12733713 Ack=1 Win=65280 Len=1448 TSval=3395800675 TSecr=3611090174
14385	7.084400890	172.16.110.1	172.16.1.10	TCP	78 54932 → 34987 [ACK] Seq=1 Ack=12735161 Win=3140480 Len=0 TSval=3611090200 TSecr=3395800675 SLE=12740953 SRE=13039241
14386	7.084515990	172.16.1.10	172.16.110.1	TCP	1514 [TCP Out-Of-Order] 34987 → 54932 [ACK] Seq=12735161 Ack=1 Win=65280 Len=1448 TSval=3395800675 TSecr=3611090174
14387	7.084523882	172.16.110.1	172.16.1.10	TCP	78 54932 → 34987 [ACK] Seq=1 Ack=12736609 Win=3139072 Len=0 TSval=3611090200 TSecr=3395800675 SLE=12740953 SRE=13039241
14388	7.084638912	172.16.1.10	172.16.110.1	TCP	1514 [TCP Retransmission] 34987 → 54932 [ACK] Seq=12736609 Ack=1 Win=65280 Len=1448 TSval=3395800675 TSecr=3611090175
14389	7.084646664	172.16.110.1	172.16.1.10	TCP	78 54932 → 34987 [ACK] Seq=1 Ack=12738057 Win=3137664 Len=0 TSval=3611090200 TSecr=3395800675 SLE=12740953 SRE=13039241
14390	7.084762253	172.16.1.10	172.16.110.1	TCP	1514 [TCP Retransmission] 34987 → 54932 [ACK] Seq=12738057 Ack=1 Win=65280 Len=1448 TSval=3395800675 TSecr=3611090175
14391	7.084770354	172.16.110.1	172.16.1.10	TCP	78 54932 → 34987 [ACK] Seq=1 Ack=12739505 Win=3136256 Len=0 TSval=3611090200 TSecr=3395800675 SLE=12740953 SRE=13039241
14392	7.084885314	172.16.1.10	172.16.110.1	TCP	1514 [TCP Retransmission] 34987 → 54932 [ACK] Seq=12739505 Ack=1 Win=65280 Len=1448 TSval=3395800675 TSecr=3611090175
14393	7.084895651	172.16.110.1	172.16.1.10	TCP	66 54932 → 34987 [ACK] Seq=1 Ack=13039241 Win=2945408 Len=0 TSval=3611090200 TSecr=3395800675
14394	7.085007817	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14395	7.085130809	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14396	7.085136676	172.16.110.1	172.16.1.10	TCP	66 54932 → 34987 [ACK] Seq=1 Ack=13042137 Win=2961920 Len=0 TSval=3611090200 TSecr=3395800675
14397	7.085254220	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14398	7.085377142	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14399	7.085384824	172.16.110.1	172.16.1.10	TCP	66 54932 → 34987 [ACK] Seq=1 Ack=13045033 Win=2993792 Len=0 TSval=3611090201 TSecr=3395800675
14400	7.085500971	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14401	7.085623404	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14402	7.085630179	172.16.110.1	172.16.1.10	TCP	66 54932 → 34987 [ACK] Seq=1 Ack=13047929 Win=3026688 Len=0 TSval=3611090201 TSecr=3395800676
14403	7.085746745	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14404	7.085869458	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14405	7.085875883	172.16.110.1	172.16.1.10	TCP	66 54932 → 34987 [ACK] Seq=1 Ack=13050825 Win=3059584 Len=0 TSval=3611090201 TSecr=3395800677
14406	7.085992729	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14407	7.086114812	172.16.1.10	172.16.110.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
14408	7.086121028	172.16.110.1	172.16.1.10	TCP	66 54932 → 34987 [ACK] Seq=1 Ack=13053721 Win=3092480 Len=0 TSval=3611090201 TSecr=3395800677

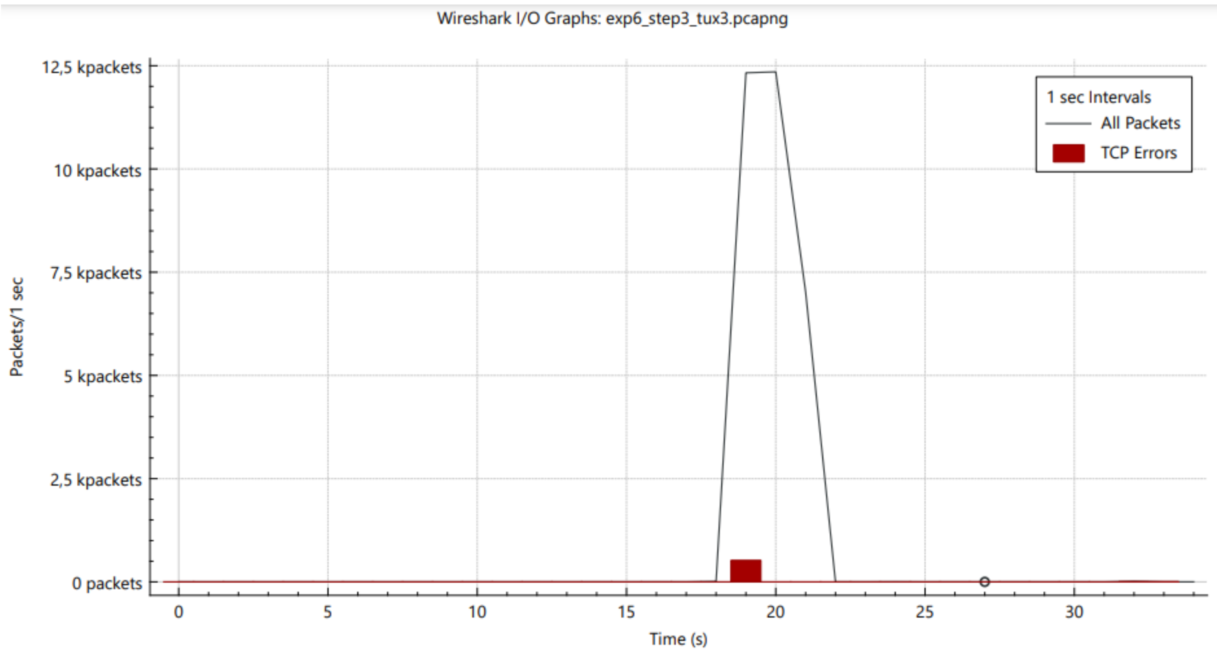
A.22 Graph of tux2 application



A.23 Graph of tux3 application step5



A.24 Graph of tux3 application step3



Annex B – Source Code

Download.h:

```
#include <stdio.h>

#include <sys/socket.h>

#include <sys/select.h>
```

```
#include <netinet/in.h>
#include <arpa/inet.h>
#include <stdlib.h>
#include <netdb.h>
#include <unistd.h>
#include <string.h>
#include <regex.h>
#include <termios.h>

#define MAX_LENGTH  500
#define FTP_PORT    21

/* Server responses */
#define SV_READY4AUTH           220
#define SV_READY4PASS           331
#define SV_LOGINSUCCESS         230
#define SV_PASSIVE              227
#define SV_READY4TRANSFER       150
#define SV_CONNECTIONALROPEN    125
#define SV_TRANSFER_COMPLETE    226
#define SV_GOODBYE              221

/* Parser regular expressions */
#define AT              "@"
#define BAR             "/"
#define HOST_REGEX      "%*[^/]//%[^/]"
#define HOST_AT_REGEX   "%*[^/]//%*[^@]@%[^/]"
#define RESOURCE_REGEX  "%*[^/]//%*[^/]/%s"
#define USER_REGEX      "%*[^/]//%[^:/]"
#define PASS_REGEX      "%*[^/]//%*[^:]:%[^@\n$]"
#define RESPCODE_REGEX  "%d"
```

```
#define PASSIVE_REGEX   "%*[^(](%d,%d,%d,%d,%d,%d)%*[^\n$)]"

/* Default login for case 'ftp://<host>/<url-path>' */
#define DEFAULT_USER        "anonymous"
#define DEFAULT_PASSWORD    "password"

/* Parser output */
struct URL {
    char host[MAX_LENGTH];      // 'ftp.up.pt'
    char resource[MAX_LENGTH];  // 'parrot/misc/canary/warrant-canary-0.txt'
    char file[MAX_LENGTH];      // 'warrant-canary-0.txt'
    char user[MAX_LENGTH];      // 'username'
    char password[MAX_LENGTH];  // 'password'
    char ip[MAX_LENGTH];        // 193.137.29.15
};

/* Machine states that receives the response from the server */
typedef enum {
    START,
    SINGLE,
    MULTIPLE,
    END,
    EXCEPTION
} ResponseState;

/*
Analisa um URL, preenche uma estrutura URL com os seus componentes (host, user, password,
resource, file, ip) e retorna 0 se não houver erro ou -1 caso contrário
*/
int parse(char *input, struct URL *url);

/*
```

Cria um descritor de arquivo de socket baseado no IP e porta do servidor fornecidos.

Retorna o descritor de arquivo do socket se não houver erro ou -1 caso contrário.

*/

int createSocket(char *ip, int port);

/*

Autentica a conexão com o servidor usando o descritor de arquivo do socket,

nome de usuário e senha fornecidos.

Retorna o código de resposta do servidor obtido pela operação.

*/

int authConn(const int socket, const char *user, const char *pass);

/*

Lê a resposta do servidor usando o descritor de arquivo do socket e preenche o buffer com a resposta do servidor.

Retorna o código de resposta do servidor obtido pela operação.

*/

int readResponse(const int socket, char *buffer);

/*

Entra em modo passivo usando o descritor de arquivo do socket e preenche ip e port com o IP e a porta da conexão de dados.

Retorna o código de resposta do servidor obtido pela operação.

*/

int passiveMode(const int socket, char* ip, int *port);

/*

Solicita um recurso ao servidor usando o descritor de arquivo do socket e o recurso desejado.

Retorna o código de resposta do servidor obtido pela operação.

*/

int requestResource(const int socket, char *resource);

```
/*

Obtém um recurso do servidor e faz o download no diretório atual usando dois descritores de arquivo de socket e o nome do arquivo desejado.

Retorna o código de resposta do servidor obtido pela operação.

*/

int getResource(const int socketA, const int socketB, char *filename);


/*

Termina a conexão com o servidor e o próprio socket usando dois descritores de arquivo de socket.

Retorna 0 se não houver erro ao fechar ou -1 caso contrário.

*/

int closeConnection(const int socketA, const int socketB);
```

Download.c:

```
#include "download.h"


int parse(char *input, struct URL *url) {


    regex_t regex;

    regcomp(&regex, BAR, 0);

    if (regexec(&regex, input, 0, NULL, 0)) return -1;


    regcomp(&regex, AT, 0);

    if (regexec(&regex, input, 0, NULL, 0) != 0) { //This case is for when there are no user credentials in the URL


        sscanf(input, HOST_REGEX, url->host);

        strcpy(url->user, DEFAULT_USER);

        strcpy(url->password, DEFAULT_PASSWORD);


    } else { // This case is for when there are user credentials in the URL (username:password@)
```

```
        sscanf(input, HOST_AT_REGEX, url->host);
        sscanf(input, USER_REGEX, url->user);
        sscanf(input, PASS_REGEX, url->password);
    }

    sscanf(input, RESOURCE_REGEX, url->resource);
    strcpy(url->file, strrchr(input, '/') + 1);

    struct hostent *h;
    if (strlen(url->host) == 0) return -1;
    if ((h = gethostbyname(url->host)) == NULL) {
        printf("Invalid hostname '%s'\n", url->host);
        exit(-1);
    }
    strcpy(url->ip, inet_ntoa(*((struct in_addr *) h->h_addr_list[0])));

    return !(strlen(url->host) && strlen(url->user) &&
           strlen(url->password) && strlen(url->resource) && strlen(url->file));
}

int createSocket(char *ip, int port) {

    int sockfd;
    struct sockaddr_in server_addr;

    bzero((char *) &server_addr, sizeof(server_addr));
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = inet_addr(ip);
    server_addr.sin_port = htons(port);

    if ((sockfd = socket(AF_INET, SOCK_STREAM, 0)) < 0) {
```

```
        perror("socket()");
        exit(-1);
    }
    if (connect(sockfd, (struct sockaddr *) &server_addr, sizeof(server_addr)) < 0) {
        perror("connect()");
        exit(-1);
    }

    return sockfd;
}

int authConn(const int socket, const char* user, const char* pass) {
    size_t userCommandLength = 5+strlen(user)+2;
    size_t passCommandLength = 5+strlen(pass)+2;
    char* userCommand = (char *)malloc(userCommandLength); sprintf(userCommand, "user %s\r\n",
user);
    char* passCommand = (char *)malloc(passCommandLength); sprintf(passCommand, "pass
%s\r\n", pass);
    char answer[MAX_LENGTH];

    write(socket, userCommand, userCommandLength);
    if (readResponse(socket, answer) != SV_READY4PASS) {
        printf("Unknown user '%s'. Abort.\n", user);
        free(userCommand);
        free(passCommand);
        exit(-1);
    }else{
        printf("Username successful\n");
    }

    if(write(socket, passCommand, passCommandLength) < 0){
        printf("Unknown password '%s'. Abort.\n", pass);
```

```
        free(userCommand);
        free(passCommand);
        exit(-1);
    }else{
        printf("Password successful\n");
    }
    return readResponse(socket, answer);
}

int passiveMode(const int socket, char *ip, int *port) {

    char answer[MAX_LENGTH];
    int ip1, ip2, ip3, ip4, port1, port2;
    write(socket, "pasv\r\n", 6);
    if (readResponse(socket, answer) != SV_PASSIVE) return -1;

    sscanf(answer, PASSIVE_REGEX, &ip1, &ip2, &ip3, &ip4, &port1, &port2);
    *port = port1 * 256 + port2;
    sprintf(ip, "%d.%d.%d.%d", ip1, ip2, ip3, ip4);

    return SV_PASSIVE;
}

int readResponse(const int socket, char* buffer) {
    char byte, exception = 0;
    int index = 0, responseCode;
    ResponseState state = START;
    memset(buffer, 0, MAX_LENGTH);

    while (state != END) {
        if (read(socket, &byte, 1) <= 0) {
```

```
    perror("read");
    return -1;
}
printf("%c", byte);

switch (state) {
    case START:
        if (byte == '-') state = MULTIPLE;
        else if (byte == ' ' && exception) {
            exception = 0;
            state = EXCEPTION;
        }else if (byte == ' ') {
            state = SINGLE;
            exception = 0;
        }else if (byte == '\n') {
            exception = 0;
            state = END;
        }else {
            buffer[index++] = byte;
            exception = 0;
        }
        break;
    case SINGLE:
        if (byte == '\n') state = END;
        else buffer[index++] = byte;
        break;
    case MULTIPLE:
        if (byte == '\n') {
            state = START;
            index = 0;
            exception = 1;
```

```
            } else {
                buffer[index++] = byte;
            }
            break;
        case END:
            break;
        case EXCEPTION:
            if (byte == '\n') state = START;
            break;
        default:
            break;
        }
    }

    sscanf(buffer, RESPCODE_REGEX, &responseCode);
    return responseCode;
}

int requestResource(const int socket, char *resource) {
    size_t fileCommandLength = 5 + strlen(resource) + 2;
    char* fileCommand = (char *)malloc(fileCommandLength);
    char answer[MAX_LENGTH];
    sprintf(fileCommand, "retr %s\r\n", resource);
    write(socket, fileCommand, fileCommandLength);
    free(fileCommand);
    return readResponse(socket, answer);
}

int getResource(const int socketA, const int socketB, char *filename) {
    FILE *fd = fopen(filename, "wb");
    if (fd == NULL) {
```

```
        printf("Error opening or creating file '%s'\n", filename);
        exit(-1);
    }

    char buffer[MAX_LENGTH];
    int bytes_read;
    int total_bytes = 0;

    while ((bytes_read = read(socketB, buffer, MAX_LENGTH)) > 0) {
        size_t bytes_written = fwrite(buffer, 1, bytes_read, fd);
        if (bytes_written != bytes_read) {
            perror("File write error");
            fclose(fd);
            return -1;
        }
        total_bytes += bytes_read;
    }

    if (bytes_read < 0) {
        perror("Socket read error");
        fclose(fd);
        return -1;
    }

    fclose(fd);

    char answer[MAX_LENGTH];
    return readResponse(socketA, answer);
}

int closeConnection(const int socketA, const int socketB) {
```

```
    char answer[MAX_LENGTH];
    write(socketA, "quit\r\n", 6);
    if(readResponse(socketA, answer) != SV_GOODBYE) {
        printf("closeConncetion error");
        return -1;
    }
    return close(socketA) || close(socketB);
}

int main(int argc, char *argv[]) {
    if (argc != 2) {
        printf("Usage: ./download ftp://[<user>:<password>@]<host>/<url-path>\n");
        exit(-1);
    }

    struct URL url;
    memset(&url, 0, sizeof(url));
    if (parse(argv[1], &url) != 0) {
        printf("Parse error. Usage: ./download ftp://[<user>:<password>@]<host>/<url-path>\n");
        exit(-1);
    }

    printf("Host: %s\nResource: %s\nFile: %s\nUser: %s\nPassword: %s\nIP Address: %s\n", url.host,
url.resource, url.file, url.user, url.password, url.ip);

    char answer[MAX_LENGTH];
    int socketA = createSocket(url.ip, FTP_PORT);
    if (socketA < 0 || readResponse(socketA, answer) != SV_READY4AUTH) {
        printf("Socket to '%s' and port %d failed\n", url.ip, FTP_PORT);
        exit(-1);
    }else{
```

```
        printf("Socket to '%s' and port %d succeeded\n", url.ip, FTP_PORT);
    }


    if (authConn(socketA, url.user, url.password) != SV_LOGINSUCCESS) {
        printf("Authentication failed with username = '%s' and password = '%s'.\n", url.user,
url.password);
        exit(-1);
    }else{
        printf("Authentication succeeded with username = '%s' and password = '%s'.\n", url.user,
url.password);
    }


    int port;
    char ip[MAX_LENGTH];
    if (passiveMode(socketA, ip, &port) != SV_PASSIVE) {
        printf("Passive mode failed\n");
        exit(-1);
    }else{
        printf("Passive mode succeeded\n");
    }


    int socketB = createSocket(ip, port);
    if (socketB < 0) {
        printf("Socket to '%s:%d' failed\n", ip, port);
        exit(-1);
    }else{
        printf("Socket to '%s:%d' success\n", ip, port);
    }


    int requestresource = requestResource(socketA, url.resource);
    if ((requestresource != SV_CONNECTIONALROPEN) && (requestresource !=
SV_READY4TRANSFER)) {
        printf("Unknown resource '%s' in '%s:%d'\n", url.resource, ip, port);
```

```
        exit(-1);
    }else{
        printf("Transfer ready to begin\n");
    }

    if (getResource(socketA, socketB, url.file) != SV_TRANSFER_COMPLETE) {
        printf("Error transferring file '%s' from '%s:%d'\n", url.file, ip, port);
        exit(-1);
    }

    if (closeConnection(socketA, socketB) != 0) {
        printf("Sockets close error\n");
        exit(-1);
    }

    return 0;
}
```